0109

National University of Computer and Emerging Sciences

Artificial Intelligence (AI 2002) Sessional-I Exam

Date: March 1, 2025

Course Instructor(s)

Dr. Asma Ahmad
Dr. Mirza Mubasher Baig
Ms. Abeeda Akram, Ms. Maham Naeem
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Total Time: 1 Hours
Total Marks: 25
Total Questions: 02

Semester: SP-2025
Campus: Lahore
Dept: Computer Science
Data Science

Student Name ______ Roll No ______ Section ______ Student Signature ______

Vetted by ______ Vetter Signature ______

WRITE THE FINAL ANSWER IN THE SPACE PROVIDED ON THE QUESTION PAPER
CLEARLY STATE ASSUMPTIONS IN CASE OF ANY MISSING INFORMATION
USE ROUGH SHEET TO PLAN YOUR ANSWER BEFORE WRITING IT

CLO 1: **Demonstrate** understanding of **basic** AI strategies for searching a large solution space

Q 1: Minimax Search [5 + 5 marks]

A program uses minimax search without pruning to play a simple game with the following rules

1. The game starts with a variable N set to an initial random value (e.g. 5).
2. Two players take turns picking a number from the set {1, 2, 3} such that the picked number must never be greater than N.
3. The chosen number is subtracted from the N.
4. The player who picks the last number (reducing N to 0) loses the game.

Assume that the initial value of N is 5 and it is programs turn to make a move.

ON THE NEXT PAGE, Draw the complete game tree used by the program to decide the move in this state and determine the move selected by the program by computing value of all nodes in the game tree.

Remember that the value of leaf nodes in this game tree will be +1 if program is winning, -1 if it is losing and 0 in case of a draw.

0109

CLO 1: **Demonstrate** understanding of **basic** AI strategies for searching a large solution space

Q 2: State-Space-Search **[4 + 2 + 7 + 2 marks]**

A computer program uses state-space-search methods to solve the following class of problems.

Given two variables, X and Y, which can hold integer values. The maximum value X can store is 4, and the maximum value Y can store is 3. Initially, both variables are set to 0.

At any step, one of the following operations can be applied on these integers:

1. **S_1:** Set X to 4 { After this operation X becomes 4}.
2. **S_2:** Set Y to 3 { After this operation Y becomes 3}.
3. **Z_1:** Set X to 0 if it is not already 0.
4. **Z_2:** Set Y to 0 if it is not already 0.
5. **T_1:** Transfer as much value as possible from X to Y without exceeding the limit of Y i.e. 3. For example if in state (3, 2) operation T_1 is applied then the new state will be (2, 3) i.e. value of X reduces by 1 and that of Y increases by 1. Similarly if T_1 is applied in state (2, 0) then the new state becomes (0, 2)
6. **T_2:** Transfer as much value as possible from Y to X without exceeding the limit of X i.e. 4. T_2 is similar to T_1 with vale transferred from Y to X.

The goal is to reach a specific target state (X, Y) using the allowed operations.

Part a. Assume that initially the variables (X, Y) are set to (0, 0) and the problem at hand is to set the values to (2, 3) i.e. X = 2 and that of Y = 3 and that the repeated states are avoided during the state-space-search

ON THE NEXT PAGE, Draw the state-space-search tree used by the program to find the solution if it used BFS to find it. Also determine the minimum and maximum numbers of states expanded by the program during the BFS search to solve this problem.

Remember we say that a state has been expanded when it comes out of the queue and its successors are inserted into the queue

Part b. What will be the minimum number of states expanded/processed by the program if it used DFS instead of BFS? Justify.

SPACE FOR TREE IN PART a

Part c. Next assume that the initial state is (4, 3) and the goal state is (2, 0). Further assume that A* search is used by the program to find the solution using the following heuristic

h(n) = Sum of Absolute Differences from the goal state,

For example, the difference of the initial state (0, 0) from the goal state (2, 0) is 2 = abs (0-2) + abs(0-0), and for a state (4,1) to the goal state (2, 0) is 3 = abs (4-2) + abs(1-0).

Draw the complete state-space-search tree used by the program to find the solution if it used A* to find it. Also write the solution found by A* and specify the states in the queue when the solution is found?

0105

Part d. Is the heuristic function, given above admissible? Justify

National University of Computer and Emerging Sciences

Artificial Intelligence (AI 2002)

Sessional-I Exam

Date: Feb 29^{th} 2024

Course Instructor(s)

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Total Time: 1 Hours

Total Marks: 35

Total Questions: **02**

Semester: SP-2024

Campus: Lahore

Dept: Computer Science
Data Science

_______________________________ _______ _______ ______________________

Student Name | Roll No | Section | Student Signature

_______________________________ ______________________

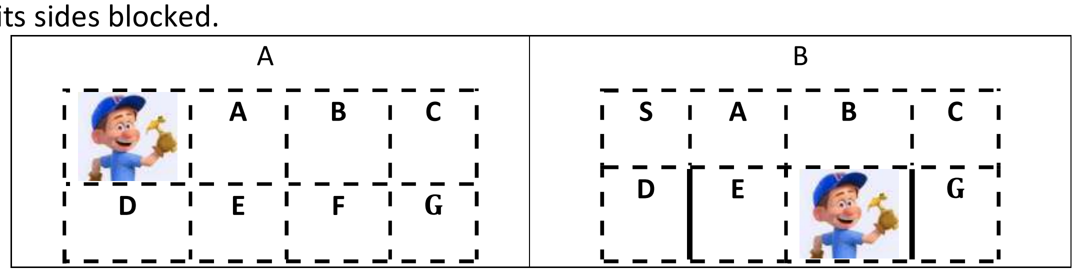
Vetted by | Vetter Signature

CAREFULLY PLAN YOUR ANSWER ON A ROUGH SHEET AND WRITE THE FINALIZED ANSWER IN THE SPACE PROVIDED ON THIS QUESTION PAPER

CLO 1: Understand principles and techniques of artificial intelligence

Q1: Search Techniques **[3 + 10 + 4 + 3 marks]**

"Solve-It Felix" is a puzzle based game played on an N x N grid of cells. An agent, **Felix**, starts from a given location on the grid and need to reach the exit cell (i.e. cell containing key). In a single move, the agent is allowed to move into either a horizontally (i.e. L, R) or vertically (i.e. U, D) adjacent cell. Some of the cell sides are permanently locked and hence cannot be used to enter into that cell. For example, in the figure (a) Felix is allowed to enter into both cells A and D whereas in figure (b) **Felix is not** allowed to enter into cell G, as that side is blocked (**SHOWN WITH SOLID LINE**) but he can enter into any other neighboring cell including cell E which also has one of its sides blocked.



For creating an auto-player for this puzzle, a teacher at NUCES-FAST proposed to use some search strategies for solving this puzzle and she needs your help in implementing the auto-player.

a) How would you represent the state of the puzzle at any instance so that it is used by the search algorithm for finding a solution?

HINT: Think about the minimum necessary information needed at any point during the execution of search algorithm for making a decision.

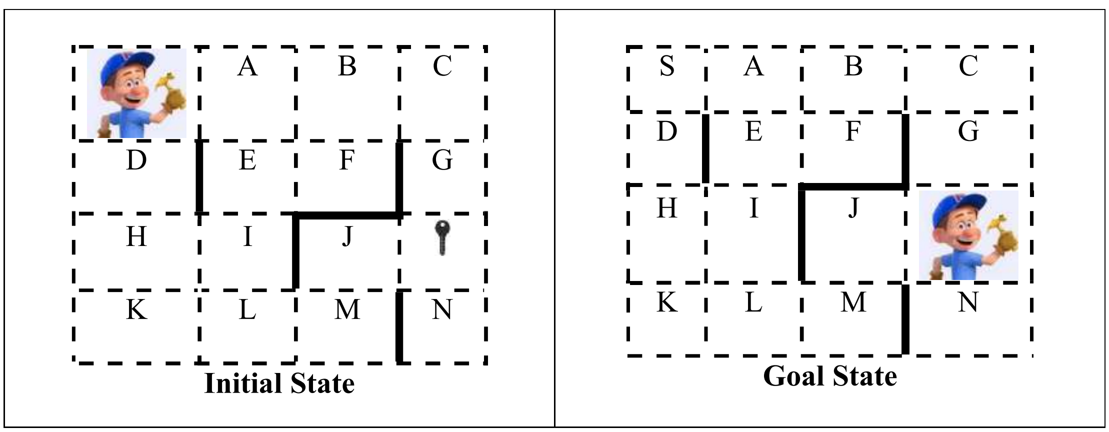
AT MINIMUM WE NEED TO KEEP TRACK OF FELIX POSITION (R, C) ON THE M X N GRID

Plus

State of the grid including position of key, sate of blocked sides might be included as these will be used to generate successors and test for the Goal state

b) Draw a complete state space search tree that will be used by the agent to plan a path from the start to the goal state assuming that implementation of the agent uses A* algorithm with the heuristic function

h(n) = Horizontal Distance of cell n from key cell + 5 * Vertical Distance of n from key cell (ignoring the blocked walls). For example the heuristic value of cell B is 1 + 5 * 2 = 11 and heuristic value of cell K is 3 + 5 * 1 = 8



Initial State | Goal State

On the tree, you must clearly mark

i. The **order** in which each node of the tree is visited/processed/expanded by A*

ii. The **value** of each node that is used by A* during processing

Also list/write the solution found by A*

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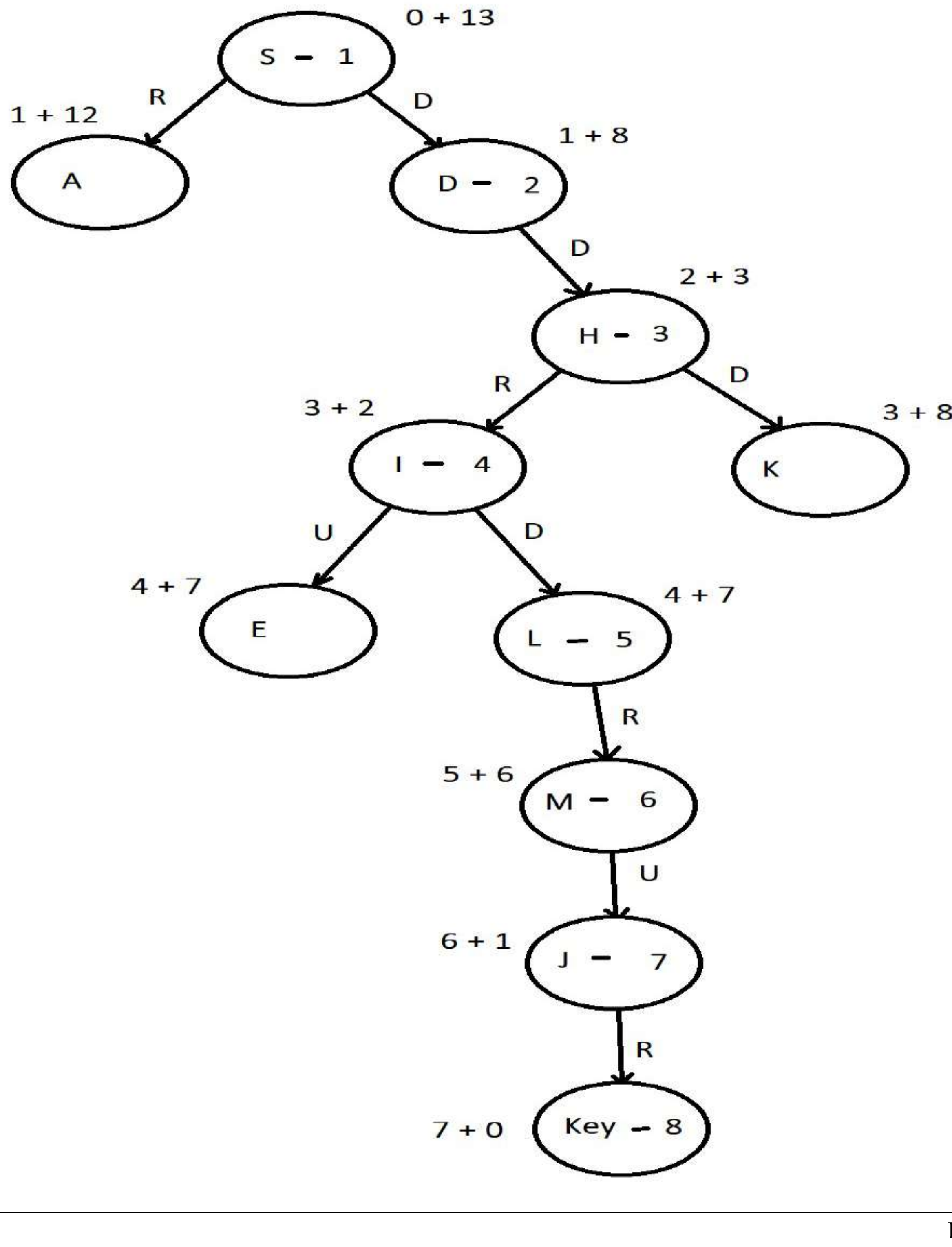
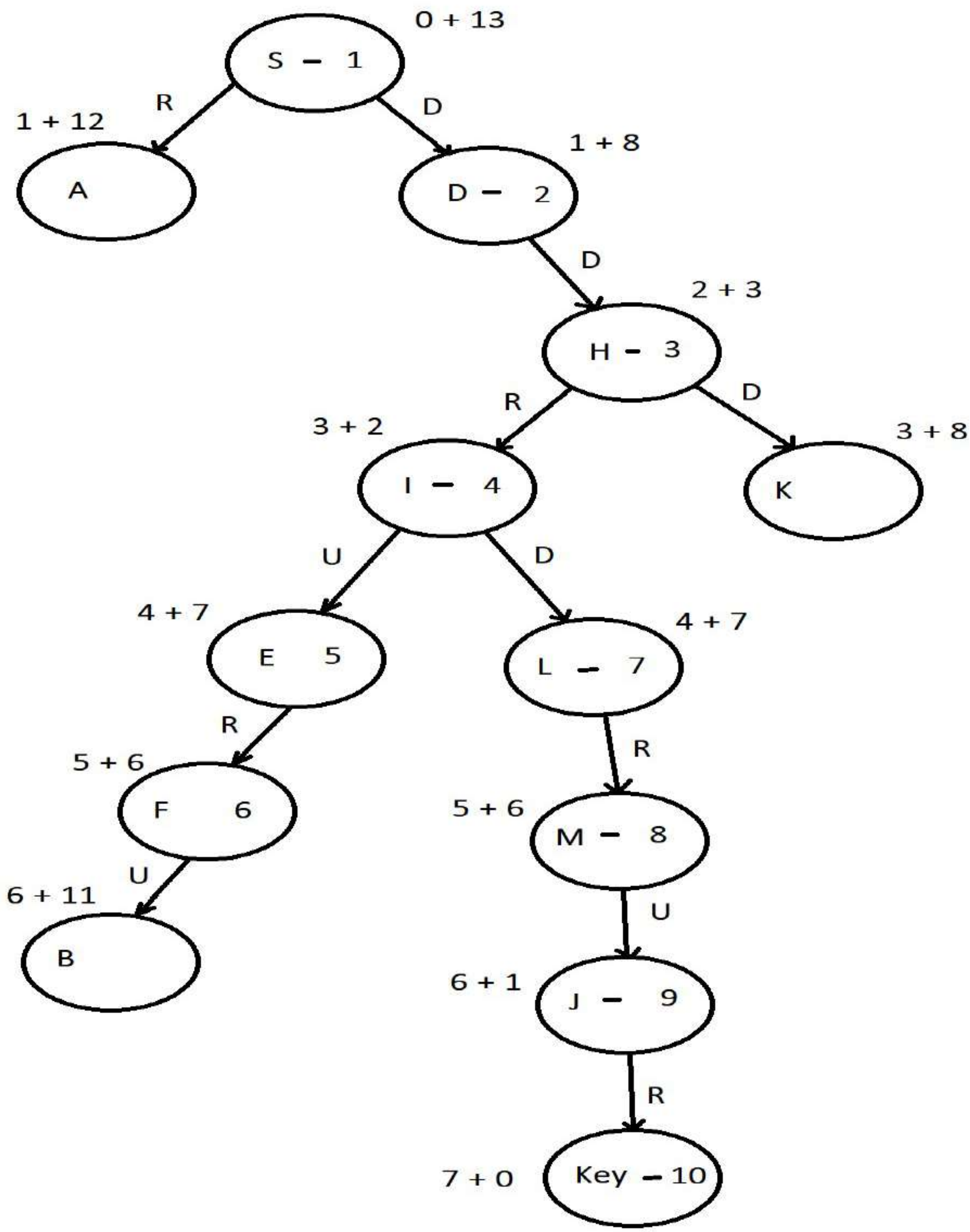
i and ii) Order of nodes expanded is given within the nodes while the cost Path + Heuristic is given outside the node. The Move is given along the arrow connecting states

The solution found will be D-D-R-D-R-U-R i.e. a total of 7 moves



A second possible tree that is a possible answer because of ties broken in slightly different way and some similar other trees with nodes expanded in a slightly different order are possible answers. However the solution found will be same if repeated nodes are avoided



c) Specify the longest (maximum number of moves) and shortest (minimum number of moves) solutions that will be generated by each of the following search strategies for the search problem of part b.

HINT: YOU DON'T NEED TO RUN ANY OF THE ALGORITHMS TO ANSWER

Search Strategy	Longest Solution	Shortest Solution
BFS	**R-R-R-D-D**	**R-R-R-D-D**
Uniform Cost Search	**R-R-R-D-D**	**R-R-R-D-D**
Iterative Deepening	**R-R-R-D-D**	**R-R-R-D-D**
A* with some admissible heuristic	**R-R-R-D-D**	**R-R-R-D-D**

All the search strategies given above give optimal solution

d)

i. Is the Heuristic function of part b admissible? Justify

THE HEURISTIC FUNCTION IS NOT ADMISSIBLE AS IT OVER ESTIMATES THE DISTANCE/MOVES FROM MANY CELLS. FOR EXAMPLE THE HEURISTIC VALUE OF G IS 5 WHEREAS THE MINIMUM COST OF G FROM KEY IS ONLY 1

CLO 1: Understand principles and techniques of artificial intelligence

Q 2: Game Playing using Minimax **[3 + 10 + 2 marks]**

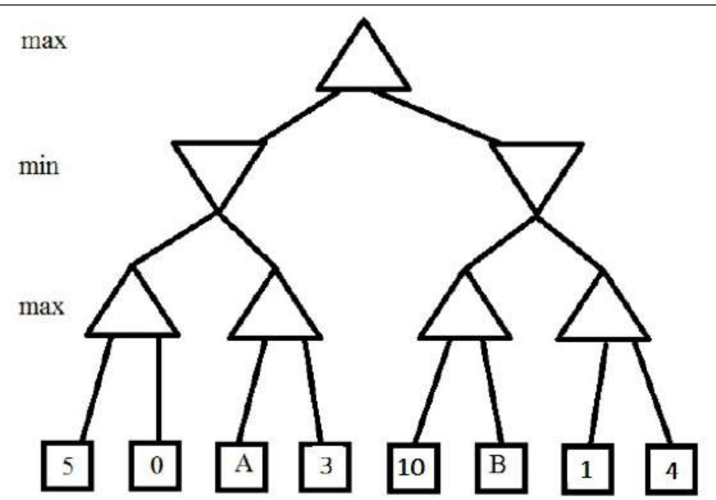
a)

For a hypothetical game tree shown on the right specify the values of A, B such that the RIGHT-SIDE move will be selected by the minimax based player

RIGHT-SIDE MOVE WILL BE SELECTED IF
MIN(5 , MAX(A,3)) < MIN (4, MAX(10, B))
i.e.
MIN(5 , MAX(A,3)) < 4

CLEARLY THE RIGHT-SIDE MOVE WILL BE SELECTED IF VALUE OF **A** IS LESS THAN 4 NO MATTER WHAT THE VALUE OF **B** IS.

Any Answer less than 4 is acceptable

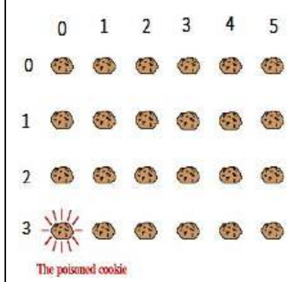


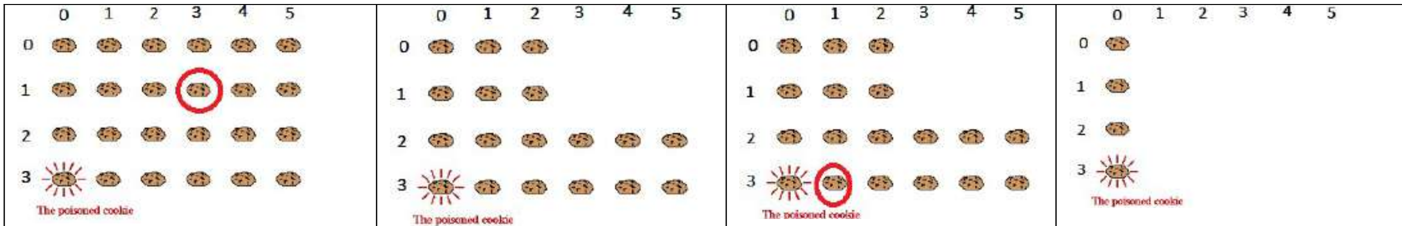
b)

Chomp is a two-player game played on a rectangular M x N grid of chocolate squares with the bottom-left square containing a poisoned bar. The figure on the right shows a 4 x 6 instance of the game

Players take turns and in each turn a player must choose a chocolate bar to eat along with all squares to the right and above it. The player who eats the poisoned square loses.

Two such possible moves along with the resulting state are shown below for the 4 x 6 grid. The encircled chocolate is selected by the player



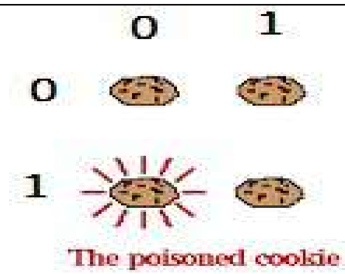


First Move	Second Move

For the 2 x 2 grid game state shown on the right

a) Built the complete game tree that will be traversed by the minimax algorithm for deciding a move at this state.

b) Compute value of each node in the game tree and hence find the move that will be selected by the player



The game tree is shown on the next page
BOX means it is a end state and hence the value of such state is either +1 or -1
Each move is shown using a tic mark
All other values can be computed using minimax with selected move shown

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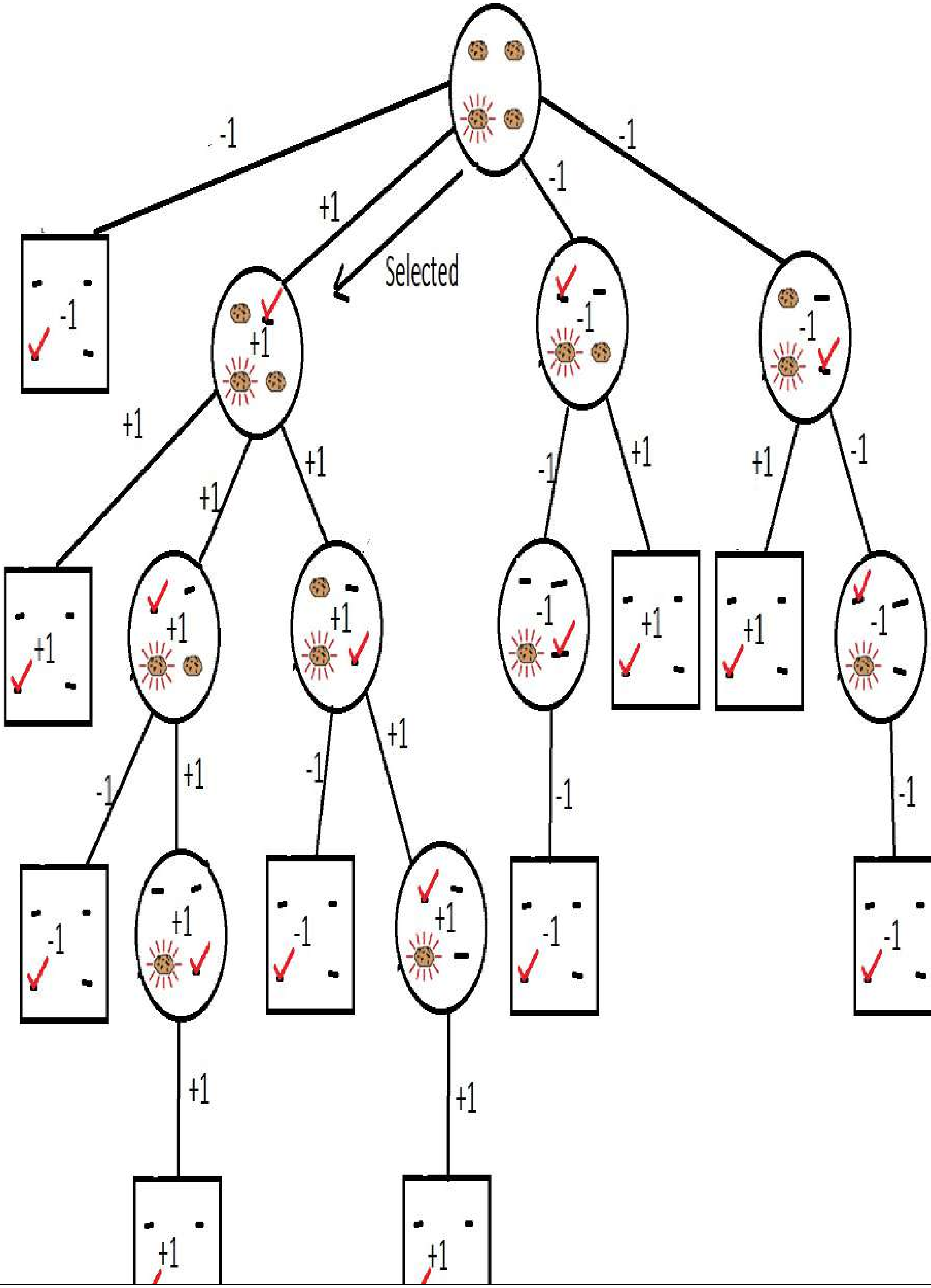


National University of Computer and Emerging Sciences, Lahore Campus

Course Name:	**Artificial Intelligence**	**Course Code:**	**AI2002**
Program:	**BS (CS) BS(DS)**	**Semester:**	**Spring 2023**
Duration:	**60 Minutes**	**Total Marks:**	**60**
Paper Date:	**27-Feb-2023**	**Weight**	**15**
Section:	**ALL**	**Page(s):**	**2**
Exam Type:	**Mid I**		

Question	**Part a** (CLO:3)	**Part b** (CLO:3)	**Part c** (CLO:2,3)	**Total Marks**
Marks	**15**	**25**	**20**	**60**
Obtained Marks				

Student Name: ________________________________ Section: ___________ Roll No.__________________

Attempt all questions. Do not use pencil or red ink to answer the questions**.** In case of confusion or ambiguity make a reasonable assumption. Use answer sheets for Part a and Part b, *but attempt **Part c** on the question paper* provided space. *Submit the question paper with answer sheets*.

QUESTION: Given below the configurations of simplified Pacman game without ghosts. The objective of the game is to eat all of the dots placed in the maze. The Pacman can move in four directions (up, down, left, and right). Whenever Pacman enters in a cell with the dot it immediately consumes it. The Grey cells in the maze are blocked.

		o	o
		o	
	o	o	

Initial State

Goal State

Part a.

Execute the state space search algorithms on the following maze where, numbers represent food cells and alphabets represents empty cells. You can build the tree with current position of Pacman instead of drawing complete states. You should create the nodes from left to right in **alphabetical order**.

	D	1	2
A		E	F
B		3	
C	5	4	

Initial State

	D		
A		E	F
B			
C			

Goal State

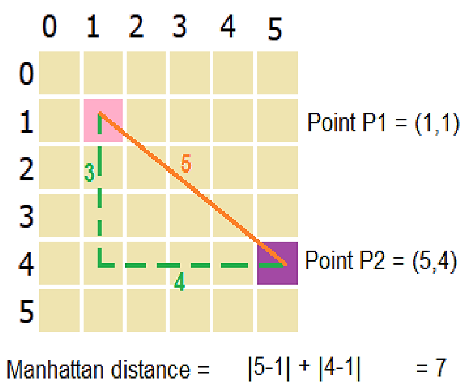
1. Depth First Search (DFS) **(Marks: 5)**
2. Breadth First Search (BFS) **(Marks: 5)**
3. Iterative Deepening Search (IDS) **(Marks: 5)**

Part b. Apply A* search on the maze provided in **Part a**. and build the search tree with the heuristic function provided below. Clearly indicate the order in which each state is expanded with open list (frontier) and closed list (visited nodes) updates.

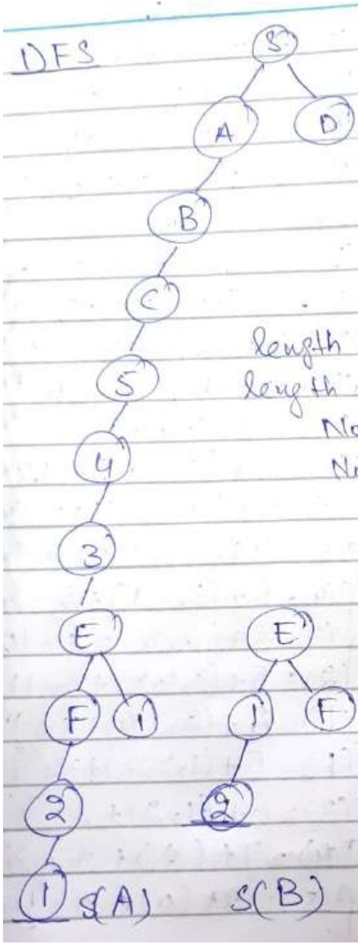
h(n) = Calculate Manhattan distance from current position to each dot and select the minimum distance + Number of remaining dots

(Marks:25)

Note: Manhattan distance will be the number of horizontal and vertical moves without considering the blocked cells.



DFS



S(A) = S→A→B→C→5→4→3→E→F→2→1

S(B) = S→A→B→C→5→4→3→E→1→2

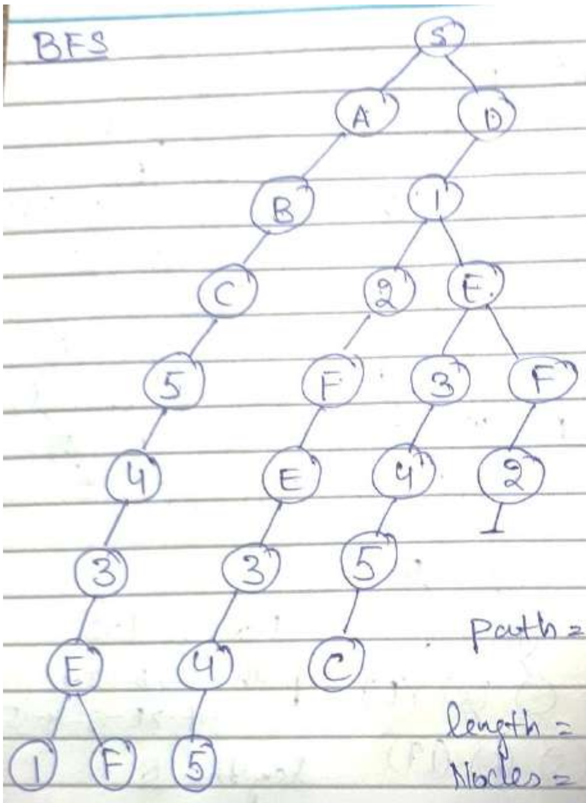
length S(A) = 10

length S(B) = 9

Nodes S(A) = 13

Nodes S(B) = 12

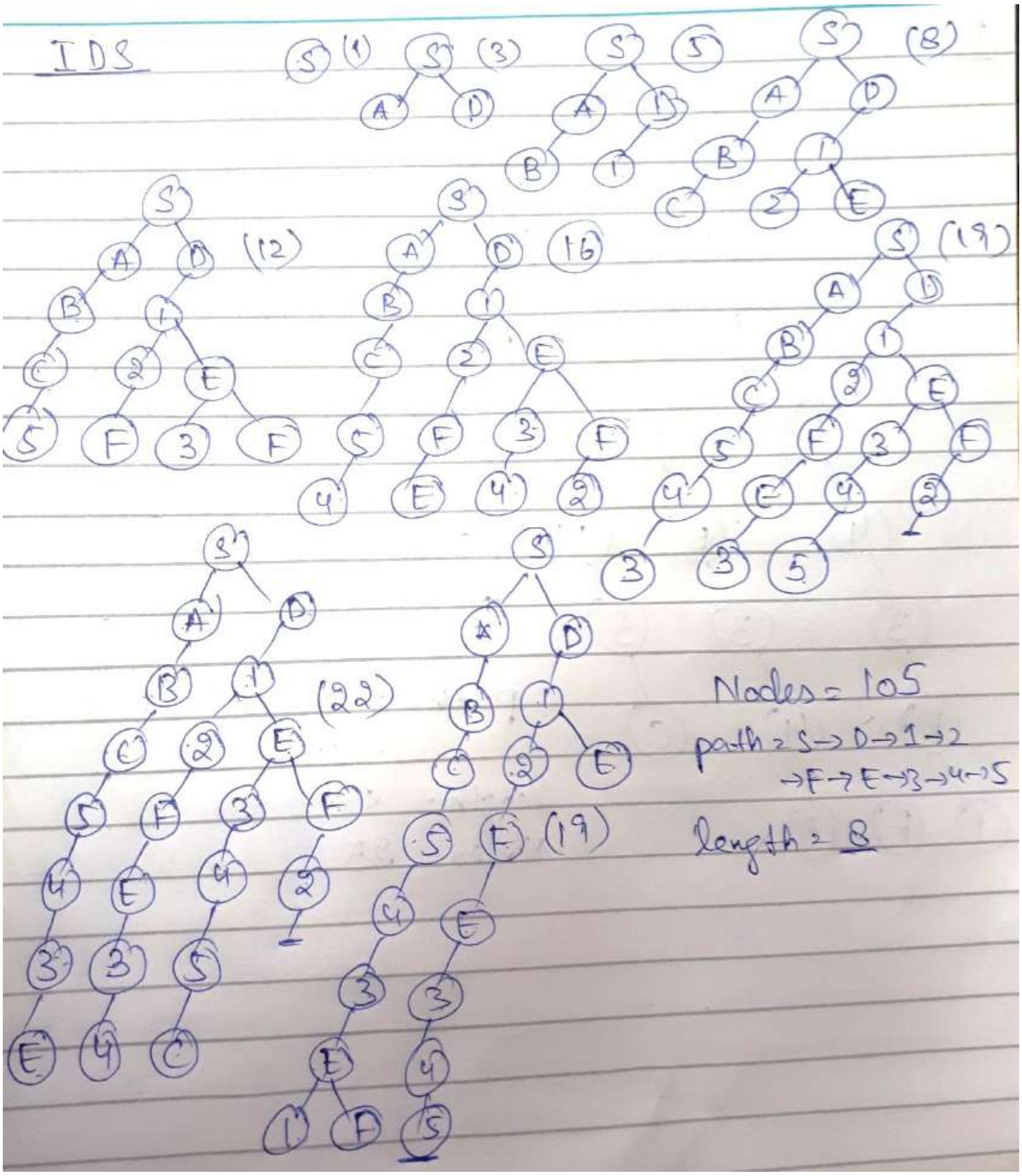
BFS



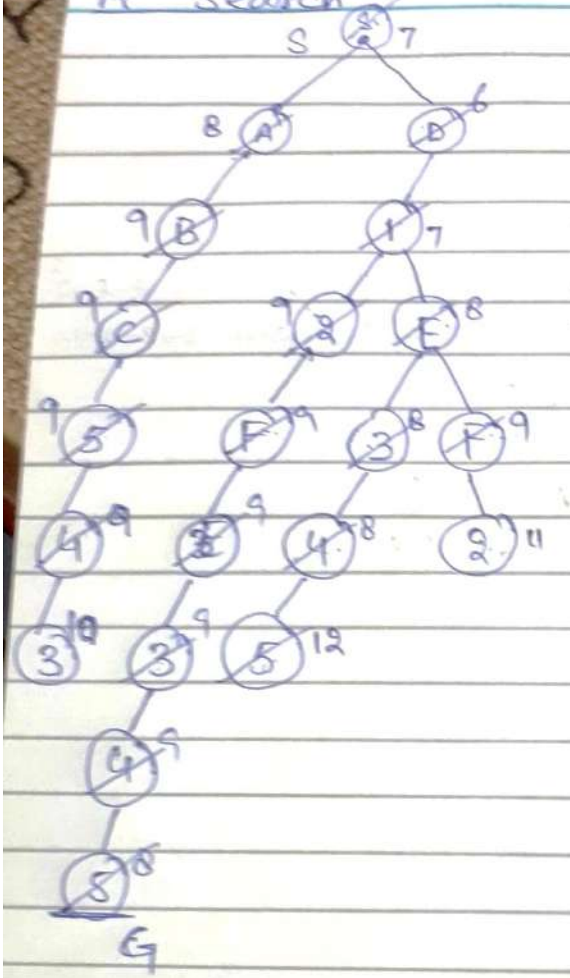
path = S→D→1→2→F→E→3→4→5

length = 8

Nodes = 25


IDS
Nodes = 105
path = S → D → 1 → 2 → F → E → 3 → 4 → 5
length = 8

A* Search



$\times V_1 \; f(S) = 0+(2,3,4,5,4)+5 = \cancel{7}$

$\times V_4 \; f(A) = 1+(3,4,3,4,3)+5 = \cancel{8}$

$\times V_2 \; f(D) = 1+(1,2,3,4,5)+5 = \cancel{6}$

$\times V_3 \; f(1) = 2+(1,3,4,5)+4 = \cancel{7}$

$\times V_{11} \; f(2) = 3+(3,4,5)+3 = \cancel{9}$

$\times V_4 \; f(E) = 3+(1,2,3)+4 = \cancel{8}$

$\times V_4 \; f(B) = 2+(4,5,2,3,2)+5 = \cancel{9}$

$\times V_5 \; f(3) = 4+(1,2)+2 = \cancel{8}$

$\times V_{10} \; f(F) = 4+(1,2,3,4)+4 = \cancel{9}$

$\times V_6 \; f(4) = 5+(1,4)+2 = \cancel{8}$

$f(5) = 6+(5)+1 = 12$

$\times V_7 \; f(C) = 3+(6,5,3,2,1)+5 = \cancel{9}$

$\times V_8 \; f(5) = 4+(5,4,2,1)+4 = \cancel{9}$

$\times V_9 \; f(4) = 5+(3,4,1)+2 = \cancel{9}$

$f(3) = 6+(2,3)+2 = 10$

$f(2) = 5+(3,4,5)+3 = 11$

$\times V_{12} \; f(F) = 4+(2,3,4)+3 = \cancel{9}$

$\times V_{13} \; f(E) = 5+(1,2,3)+3 = \cancel{9}$

$\times V_{14} \; f(3) = 6+(1,2)+2 = \cancel{8}$

$\times V_{15} \; f(4) = 7+(1)+1 = \cancel{8}$

$\times V_{16} \; f(5) = 8+(0)+0 = \cancel{8}$

$S \to D \to 1 \to 2 \to F, E, 3 \to 4 \to 5$

path length = 8

optimal = yes

Nodes = 21

Part c.

1. Which of the following searches are Incomplete? **(Marks:3)**
    a) Iterative deepening search
    b) Breadth first search
    c) **Greedy best first search with admissible heuristics**
    d) All of the above
2. Which of the following searches are optimal in case all actions have equal cost. **(Marks:3)**
    a) Uniform cost search
    b) Breadth first search
    c) A* search with admissible heuristics
    d) **All of the above**
3. What is the difference between Completeness and Optimality? **(Marks:4)**

    Completeness - does the search always find a solution if one exists?

    Optimality - does it always find a least-cost solution?

4. Heuristic function used in **Part b** is admissible or not? Why? **(Marks:4)**

    Yes, the heuristic function is admissible because the actual cost to reach a dot is greater than or equal to the estimated heuristic value of H(n). Actual cost is greater because it incorporates the cost of hurdles placed in the path but H(n) is always underestimating this value.

5. Provide the comparison of all search strategies used for Pacman in the following table. **(Marks:8)**

Algorithms	DFS	BFS	IDS	A*
Total No. of Nodes	13/12	25	105	21
Solution Path Found	SABC543EF21/ SABC543E12	SD12FE345	SD12FE345	SD12FE345
Solution Path Length	10/9	8	8	8
Optimal	No/No	Yes	Yes	Yes

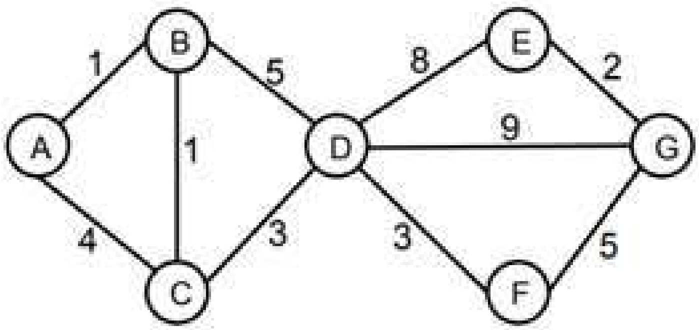
Roll No.________________

National University of Computer and Emerging Sciences, Lahore Campus

Course Name:	Artificial Intelligence	Course Code:	
Program:		Semester:	Spring 2020
Duration:	60 Minutes	Total Points:	
Paper Date:	Thursday, March 24, 2022	Weight	
Section:	ALL	Page(s):	
Exam Type:	Mid-I		

Student : Name:______________________________ Roll No.________________ Section:_______

Problem. Search Algorithms [5 + 5 Points]



Node	h_1	h_2
A	9.5	10
B	9	12
C	8	10
D	7	8
E	1.5	1
F	4	4.5
G	0	0

Consider the state space graph shown above. **A** is the start state and **G** is the goal state. The costs for each edge are shown on the graph. Each edge can be traversed in both directions.

Part a) For each of the following graph search strategies, mark which, if any, of the listed paths it could return. Note that for some search strategies the specific path returned might depend on tie-breaking behavior. In any such cases, make sure to mark all paths that could be returned under some tie-breaking scheme. **[5 Points]**

	A-B-D-G	A-C-D-G	A-B-C-D-F-G
Depth first search	✓	✓	✓
Breadth first search	✓	✓	
Uniform cost search			✓
A* search with heuristic h_1			✓
A* search with heuristic h_2			✓

Roll No.________________

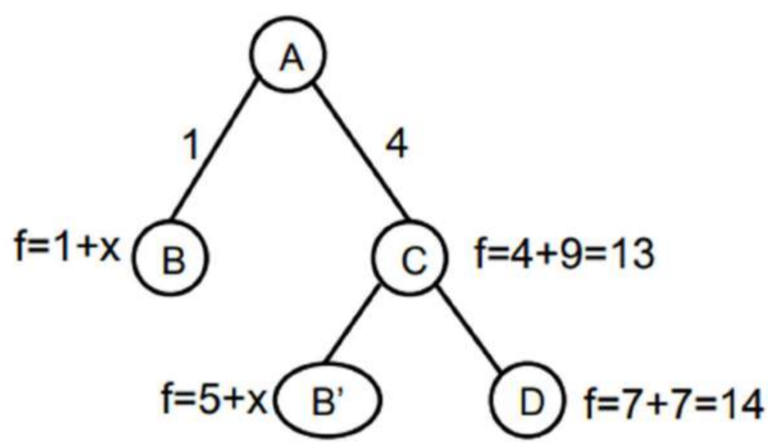
Part b) Suppose you are completing the new heuristic function $\mathbf{h_3}$ shown below. All the values are fixed except $\mathbf{h_3(B)}$. For each of the following conditions, write the set of all values that are possible for $\mathbf{h_3(B)}$. **[2 + 3 Points]**

Node	A	B	C	D	E	F	G
h_3	10	?	9	7	1.5	4.5	0

i. What values of $h_3(B)$ make h_3 admissible? Justify your answer.

Any value in the range [0 12] because the value must be less than or equal to the optimal cost which is 12

ii. What values of $h_3(B)$ will cause A* graph search to expand node A, then node C, then node B, then node D in order? Justify your answer.



If $h_3(B) = x$ then

If NODE C is to expanded before NODE B and after NODE A then 1 + x must be greater than 13 so $1 + x > 13$ implies $x > 12$

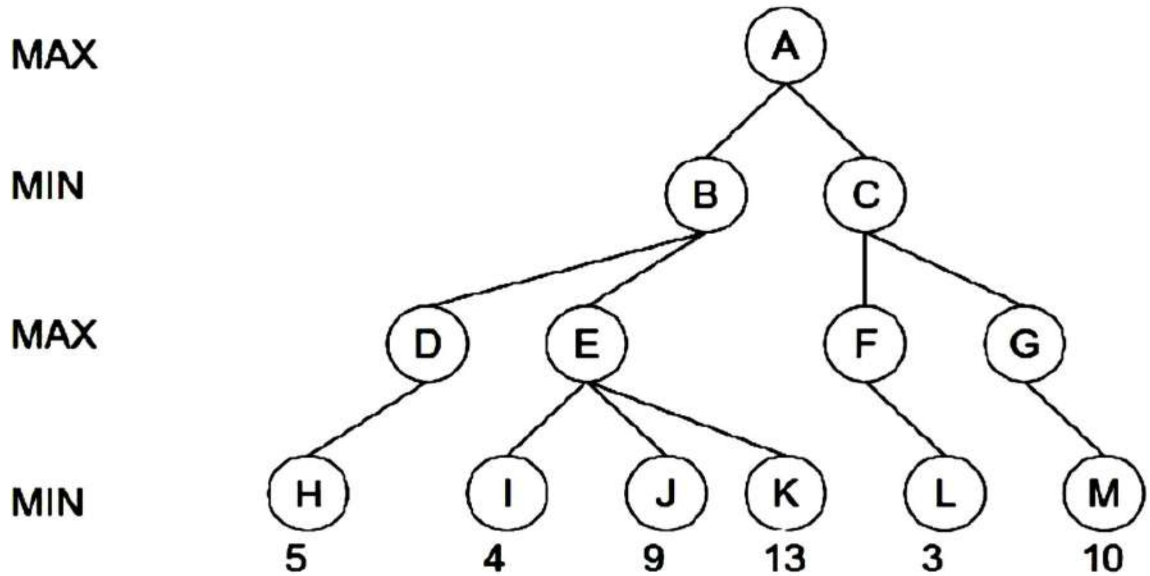
Similarly if NODE B is to be processed before NODE D then 1 + x must be smaller than 14 hence $1 + x < 14$ implies $x < 13$

Therefore any value of $h_3(B)$ in the range (12 13) will cause the nodes to expand in the desired order

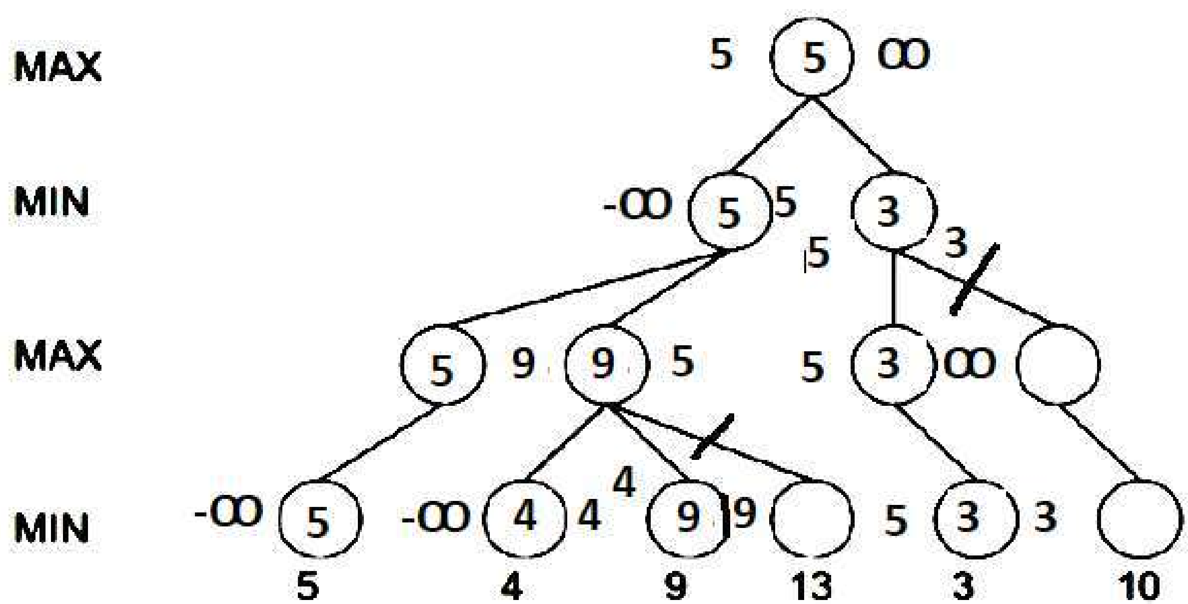
Roll No.________________

Problem. Adversarial Search [4 + 3 + 3 Points]

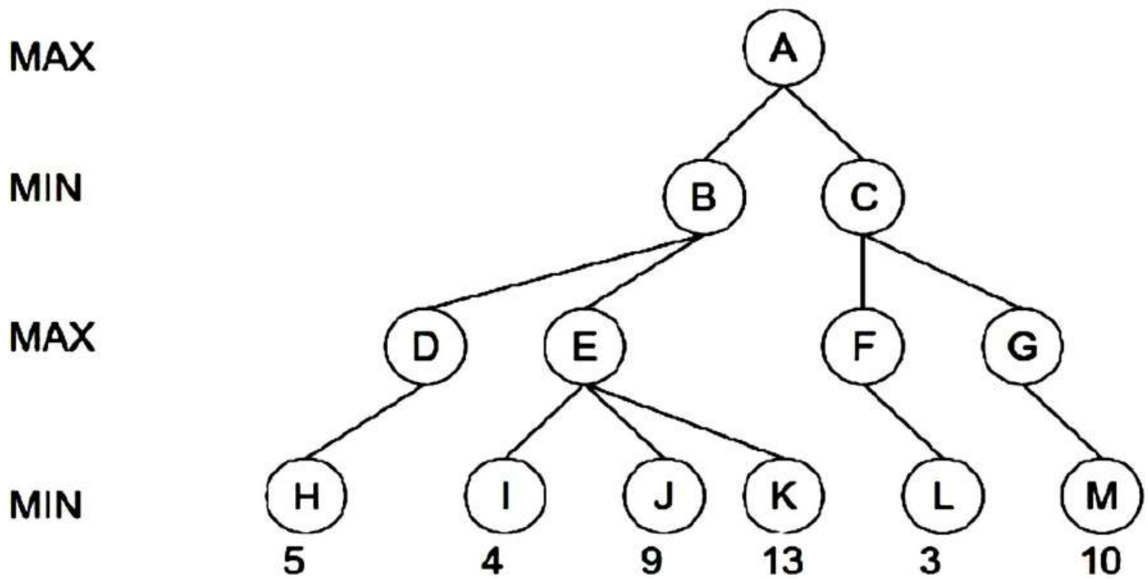
Consider the game tree shown below with the value of each node at level 3 computed using ꝏꝏa naïve evaluation function. Assume the nodes are explored from **left to right** and minimax with standard alpha beta pruning is used.



Part a) Write on the tree the **value of each node** as computed by minimax with alpha-beta pruning. Also mark each branch that will be pruned during the search. **[4 Points]**



Part b) On the figure shown below, mark the branches pruned if the nodes are expanded from right to left instead of left to right **[3 Points]**

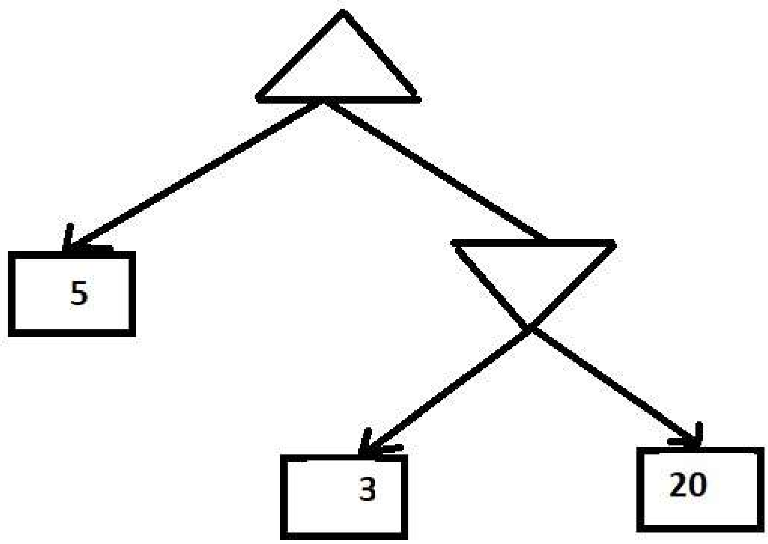


For this case no branches will be pruned

Roll No.________________

Part c) Disprove the following statement about minimax by providing a counter example and then write the correct version of this statement. **[3 Points]**

Statement: Minimax always results in an optimal decision against any player.



For the above game tree the minimax will return the left move with a gain of 5 whereas going right can result in a better gain if the opponent is not perfect

Correct Statement: Minimax always results in an optimal decision against a **perfect** player

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National University of Computer and Emerging Sciences, Lahore Campus

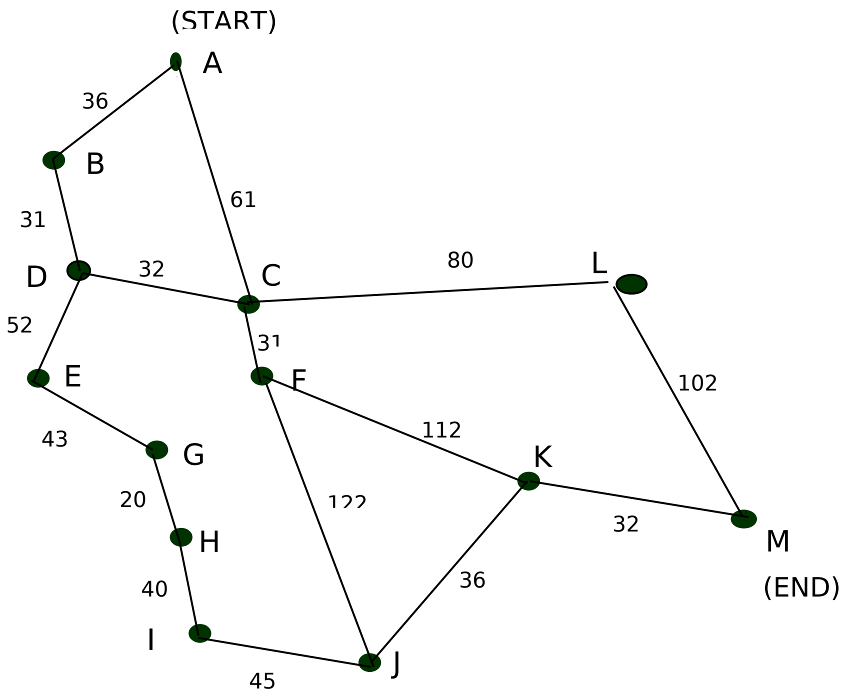
Course Name:	Artificial Intelligence	Course Code:	CS 401
Program:	BS (Computer Science)	Semester:	Spring 2020
Duration:	60 Minutes	Total Marks:	75
Paper Date:	25-2-2020	Weight	15
Section:	All	Page(s):	6
Exam Type:	Mid I		

Student Name:
Registration #: Section:

Instructions: No rough sheets needed.

Q1. A* Search Algorithm: (25)

Consider the following map.



Using the A* algorithm avoiding repeated states, work out a route from town A to town M.

Use the following cost functions.

- g (n) = The cost of each move as the distance between each town (shown on map).
- h (n) = The Straight Line Distance between any town and town M. These distances are given in the table below.

Provide the search tree for your solution and indicate the order in which you expanded the nodes. Finally, state the route you would take and the cost of that route.

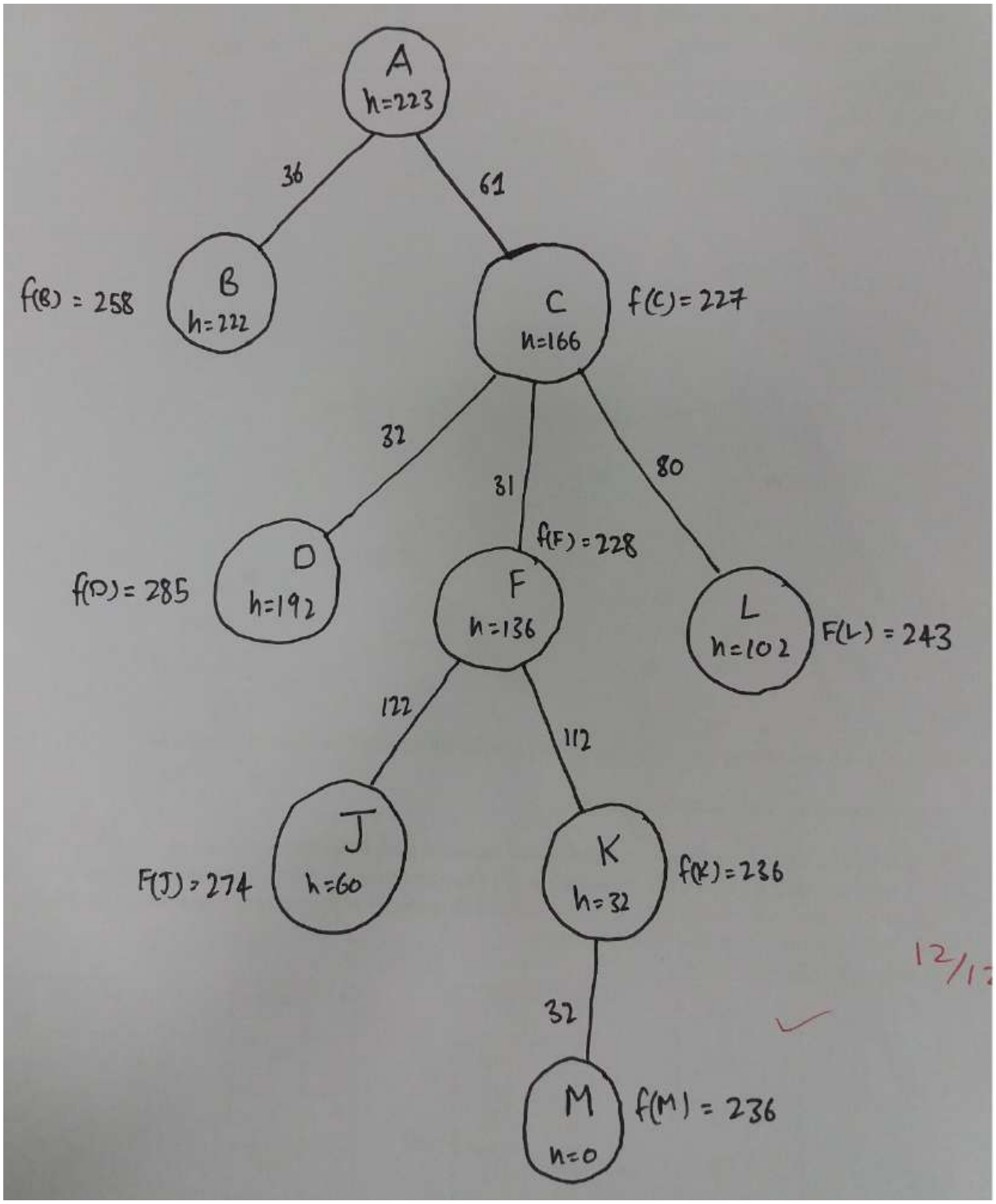
<u>Straight Line Distance to M</u>

A	223	E	165	I	100	M	0
B	222	F	136	J	60		
C	166	G	122	K	32		

D	192	H	111	L	102

In your answer provide the following:

i) The search tree that is produced, showing the cost function at each node (12 marks)



ii) State the order in which the nodes were expanded (5 marks)

ACFKM

State the route that is taken, and give the total cost (4 mark)

ACFKM

Explain the relationship between the A* algorithm and the Uniform Cost Search algorithm? (4 marks)

A*: f(n) = g(n) + h(n)
Uniform Cost Search = f(n) = g(n)

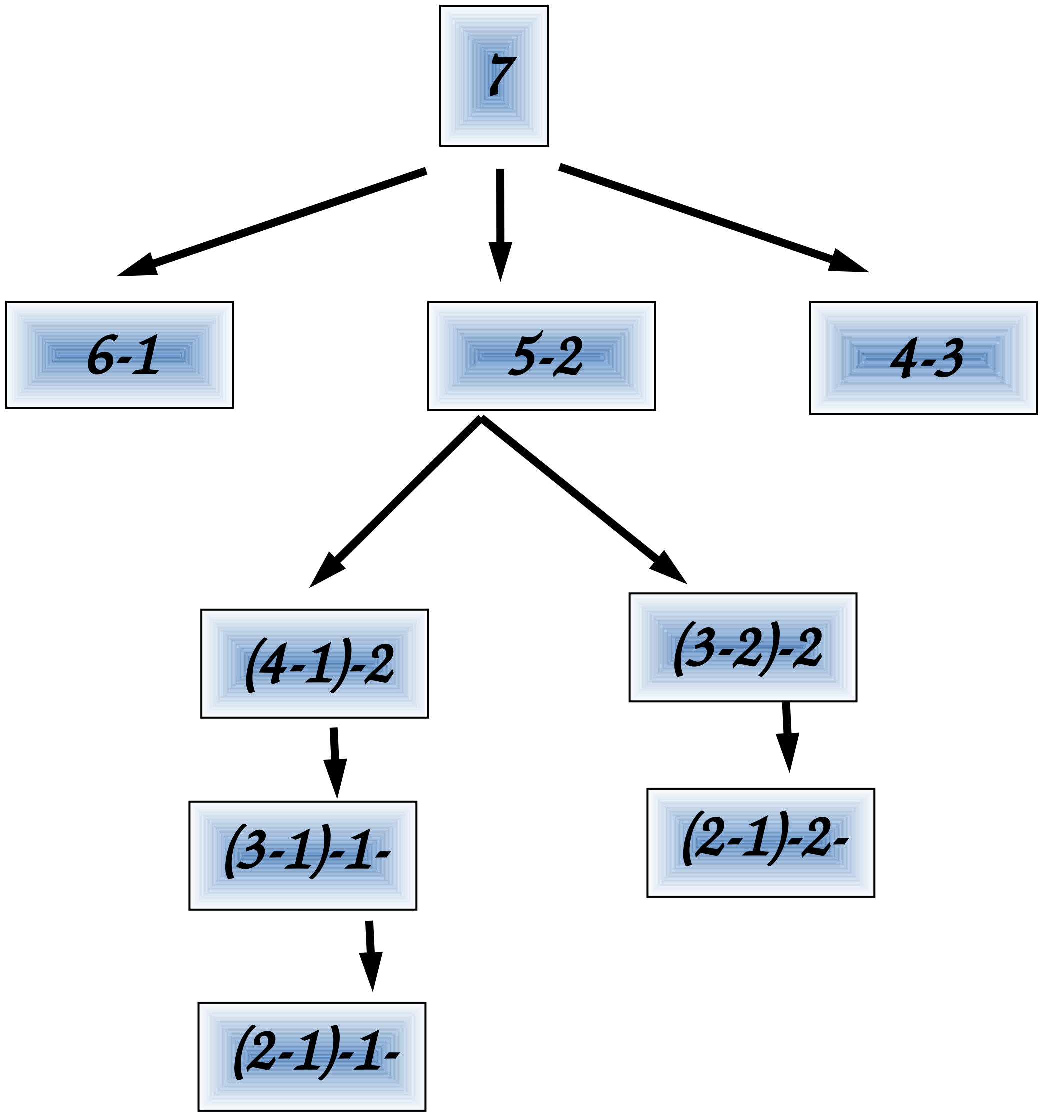
Reference:
When all step costs are equal, breadth-first search is optimal because it always expands the shallowest unexpanded node. By a simple extension, we can find an algorithm that is optimal with any step-cost function. Instead of expanding the shallowest node, uniform-cost search expands the node n with the lowest path cost g(n).

Q2: Nim is a two-player game The rules are as follows. (25)

The game starts with a single pile of 7 tokens. At each move a player selects one stack and divides it into two non-empty, non-equal piles. A player who is unable to move loses the game.

a. Moving first a human player chose to divide the pile of 7 tokens into a two piles of sizes 5 and 2 respectively as shown below. It is turn of auto-player. Draw the complete search tree for the auto-player.

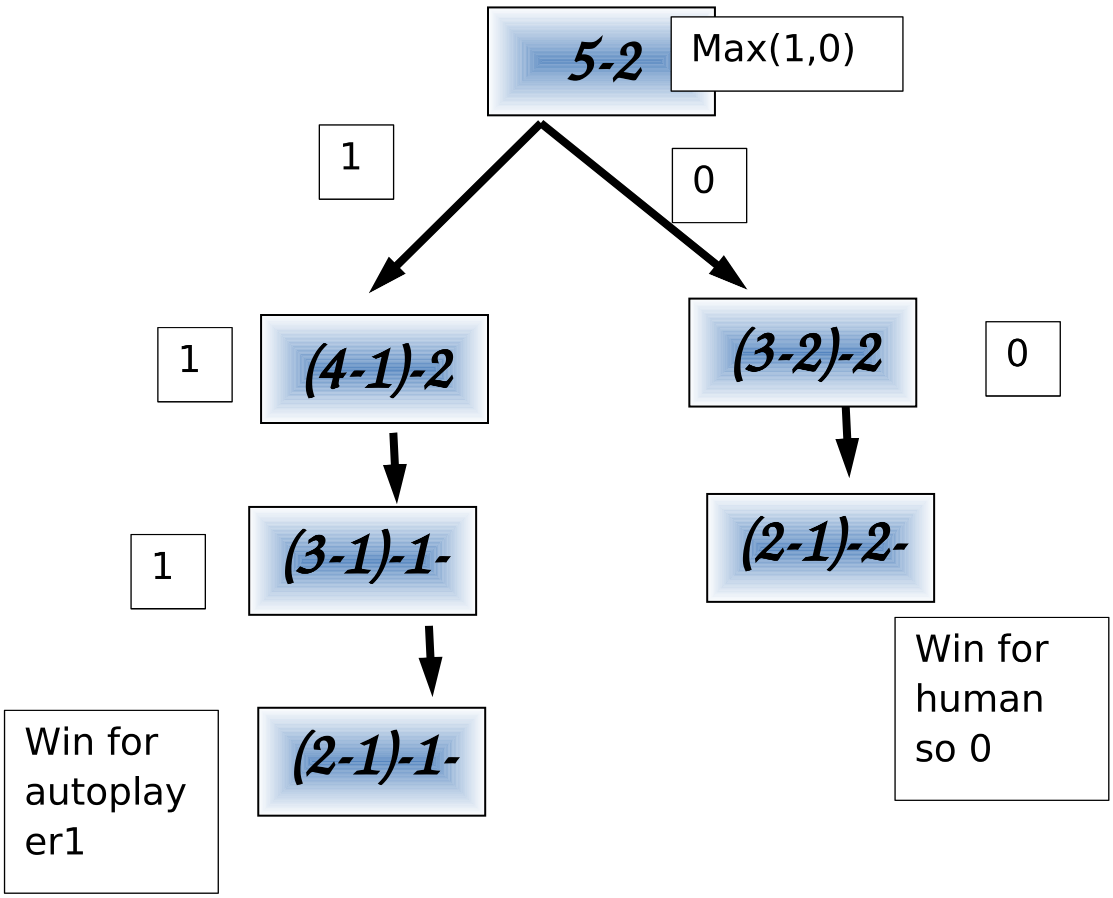
b.



(b) Assume two players, min and max, play nim (as described above). If a terminal state in the search tree developed above is a win for min, a utility function of zero is assigned to that state. A utility function of 1 is assigned to a state if max wins the game.

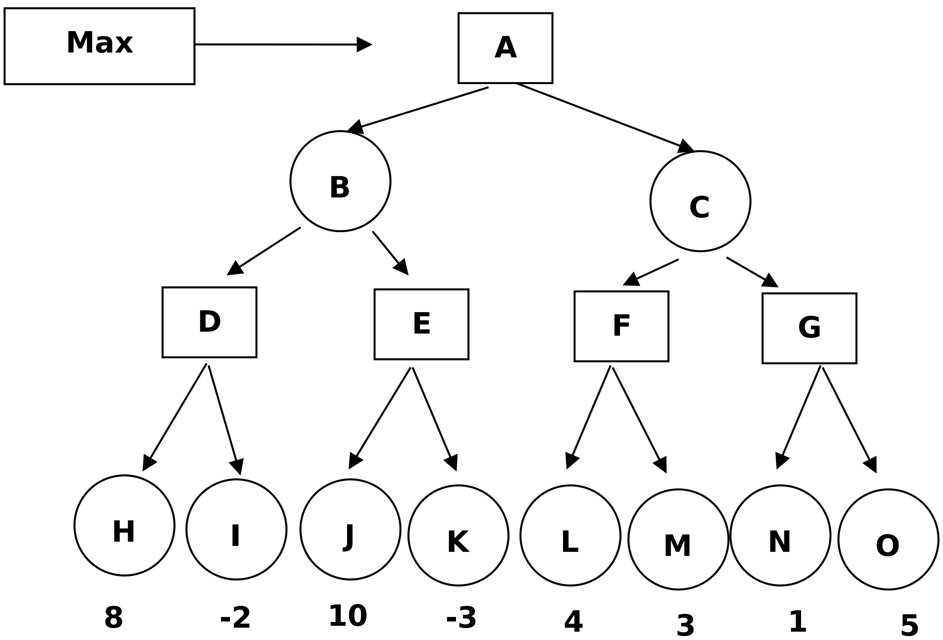
Apply the minimax algorithm to the tree drawn in pervious question and decide the move auto-player must make.

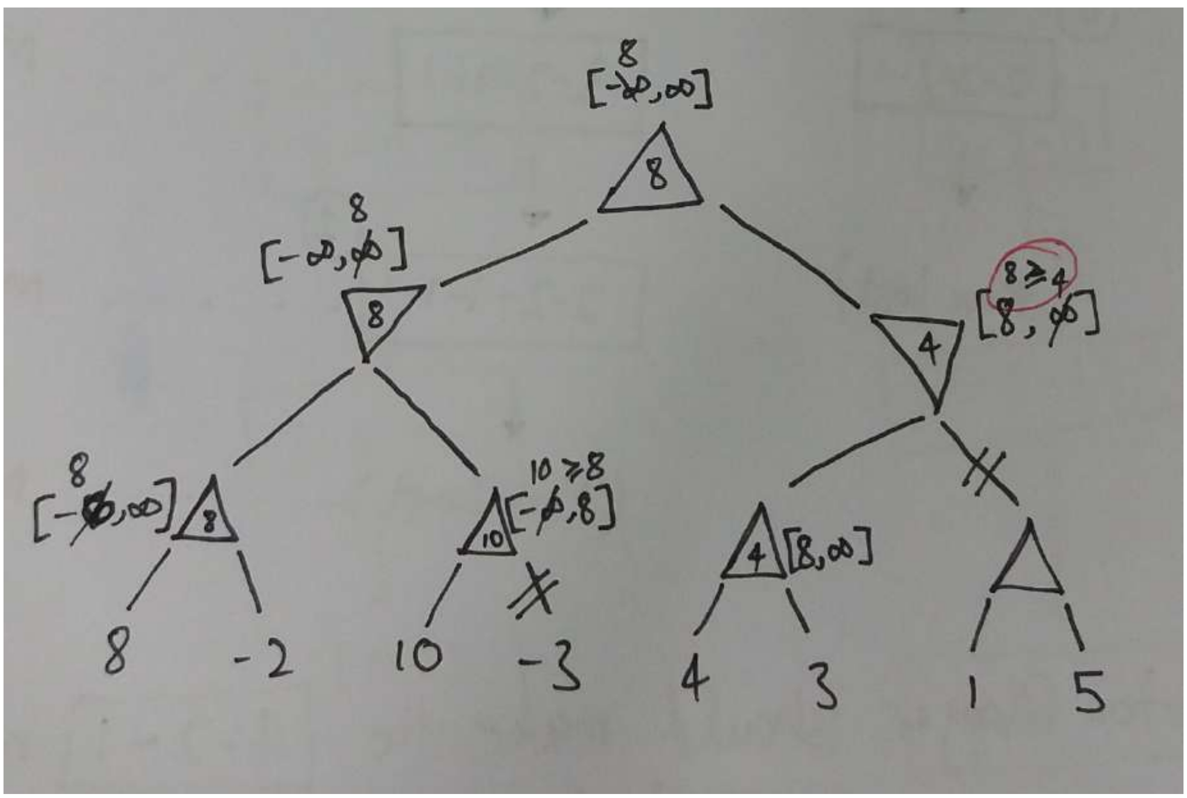
Assuming computer is max



Q3: Given the following search tree, apply the alpha-beta pruning algorithm to it and show the move search tree that would be built by this algorithm. Make sure that you show where the alpha and beta cuts are applied and which parts of the search tree are pruned as a result. Assume the nodes are explored from **left to right**.

(25)





Should be 8>=8

Good Luck

National University of Computer and Emerging Sciences, Lahore Campus

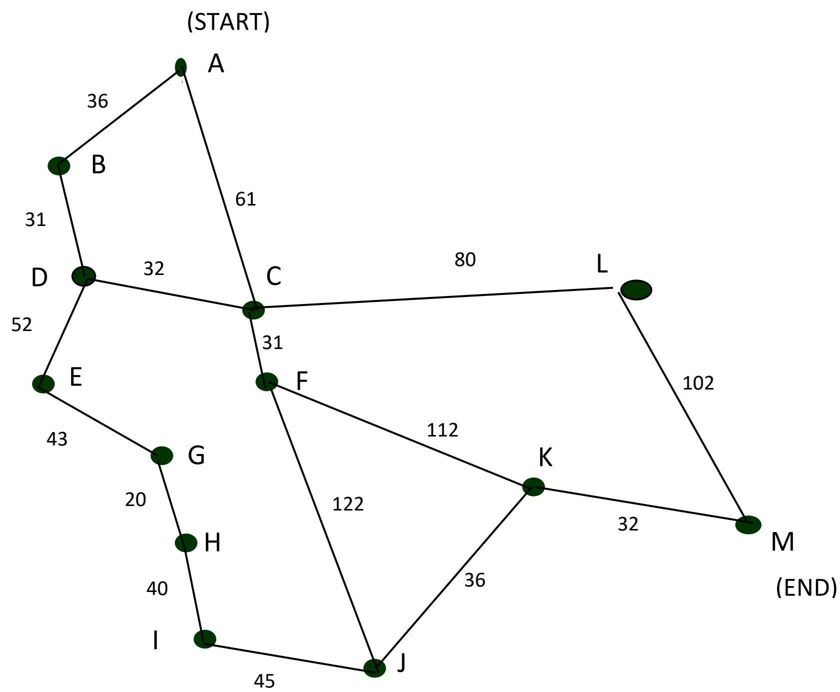
Course Name:	**Artificial Intelligence**	**Course Code:**	**CS 401**
Program:	**BS (Computer Science)**	**Semester:**	**Spring 2020**
Duration:	**60 Minutes**	**Total Marks:**	**75**
Paper Date:	**25-2-2020**	**Weight**	**15**
Section:	**All**	**Page(s):**	**6**
Exam Type:	**Mid I**		

Student Name:
Registration #: Section:

Instructions: No rough sheets needed.

Q1. A* Search Algorithm: (25)

Consider the following map.



Using the A* algorithm avoiding repeated states, work out a route from town A to town M.

Use the following cost functions.

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Provide the search tree for your solution and indicate the order in which you expanded the nodes. Finally, state the route you would take and the cost of that route.

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In your answer provide the following:

i) The search tree that is produced, showing the cost function at each node (12 marks)

ii) State the order in which the nodes were expanded (5 marks)

State the route that is taken, and give the total cost (4 mark)

Explain the relationship between the A* algorithm and the Uniform Cost Search algorithm? (4 marks)

Q2: Nim is a two-player game The rules are as follows. (25)

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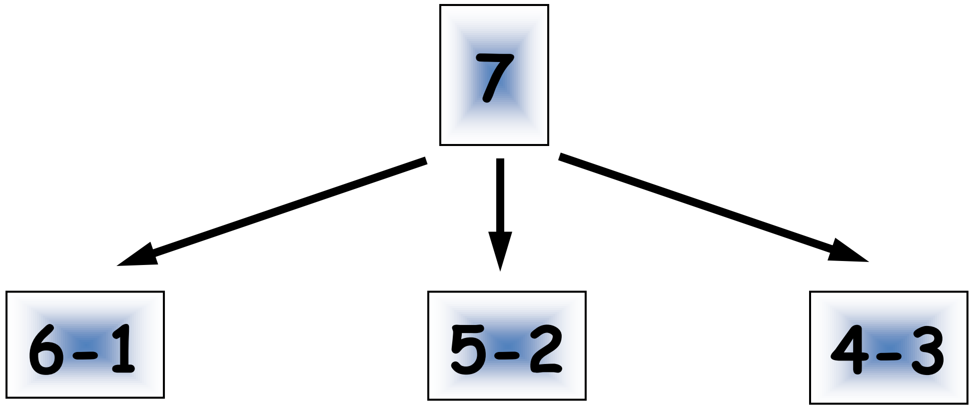

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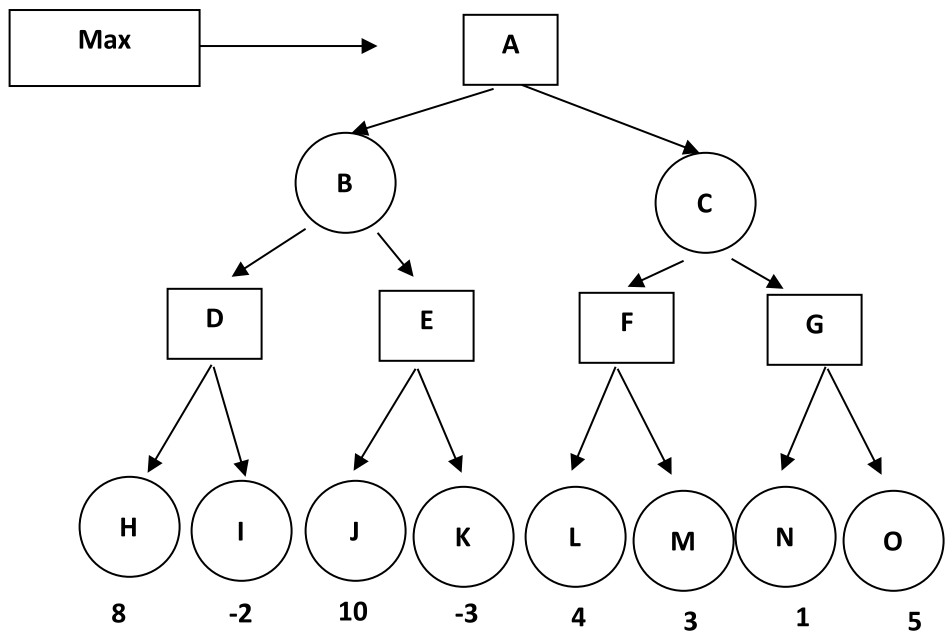


Good Luck

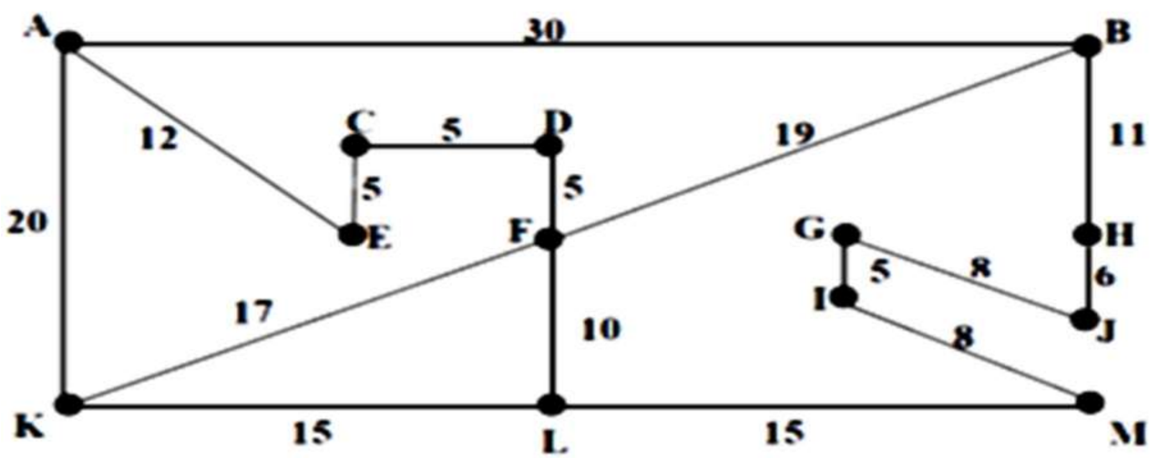
National University of Computer and Emerging Sciences, Lahore Campus

Course Name:	Artificial Intelligence	Course Code:	CS 401
Program:	BS (Computer Science)	Semester:	Spring 2019
Duration:	60 Minutes	Total Marks:	50
Paper Date:	26-2-2019	Weight	15
Section:	For all sections	Page(s):	6
Exam Type:	Mid I		

Student Name:
Registration #:

Q1. A* Search Algorithm: **(10 + 8 + 2 Marks)**

Consider the following map (not drawn to scale)



Part a) Use A* algorithm to work out a route from town A to town M. Use the following cost functions

g(n) = Total cost of reaching from town A to town n (step cost of each move is given on the map)

h(n) = Straight line distance between town n and town M These costs are given in the table below.
h(n) can be used as an estimated distance to M

Town	Distance
A	56
B	22
C	30
D	29
E	29
F	30
G	14

Town	Distance
H	10
I	8
J	5
K	30
L	15
M	0

i) Showing the order of nodes expanded by A* (Show complete working on the next page)

ii) What path/route would be found by A* using this heuristic function?

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Part b) Repeat the above question for the following heuristic function

Town	Distance
A	80
B	10
C	50
D	20
E	10
F	30
G	60

Town	Distance
H	30
I	20
J	50
K	60
L	20
M	0

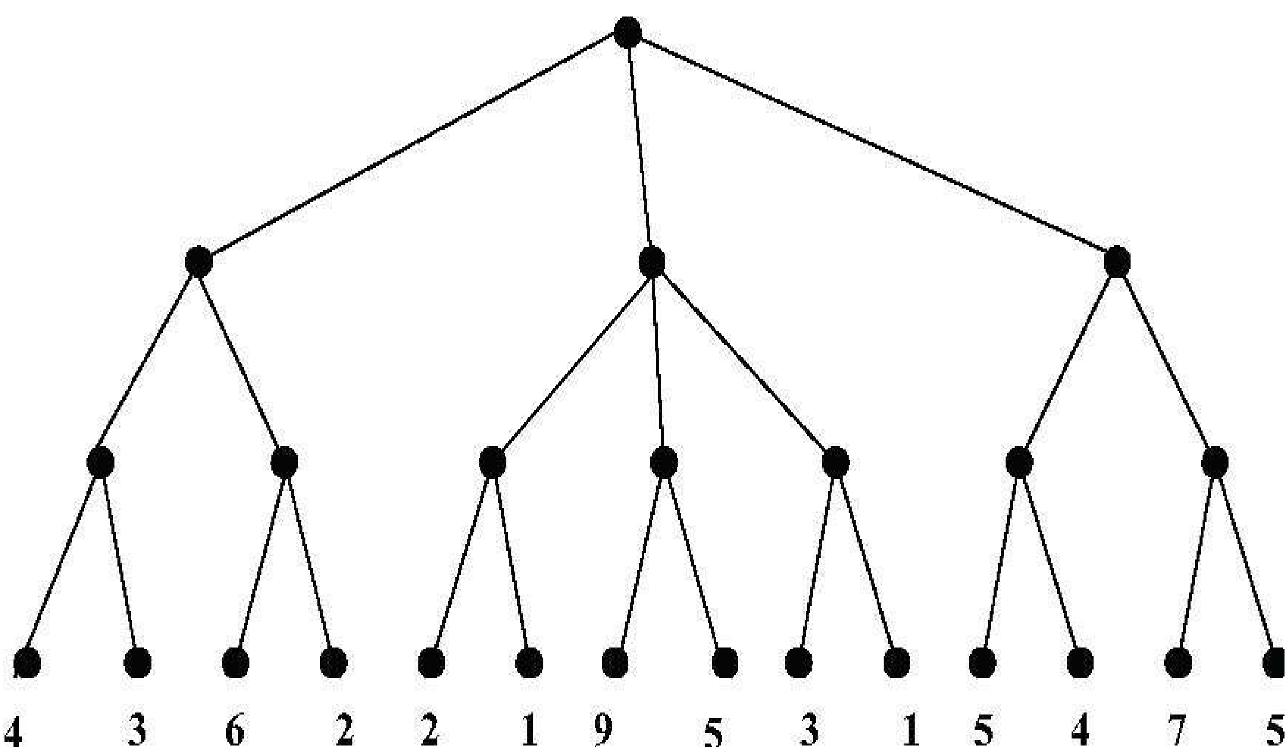
Part c) Comment on the optimality of the two runs of A* algorithm in the first two parts.

How would you account for the different routes returned by the two runs?

Q2. Run MINIMAX algorithm with Alpha-Beta pruning on the given game tree. Start state is for MAX and the values given at the bottom are the values of corresponding states for max player. **(8 + 7 mark).**

On the figure clearly show

- value of each node as found by the minimax algorithm
- the branches that will be pruned by the algorithm and the values of alpha and beta at the time of pruning



Q3. While solving eight puzzle we want to reach the state shown on right (**GOAL**) from the state shown on the left (**INITIAL STATE**) **[5 + 5 + 5 Marks]**

Initial State

2	8	3
1		4
7	6	5

Goal State

1	2	3
8		4
7	6	5

a) In the form of a tree, show all states processed by iterative deepening algorithm while finding the solution.
b) Specify the path found by iterative deepening algorithm.

c) Also show the number of times each node is processed/expanded by iterative deepening.

You must assume that the basic DFS has been implemented such that it always process the four possible moves in LEFT, UP, RIGHT and DOWN order i.e. the node generated by moving the space down is pushed onto the stack first followed by the node generated by moving the space right and so on.

Blank Page

Good Luck

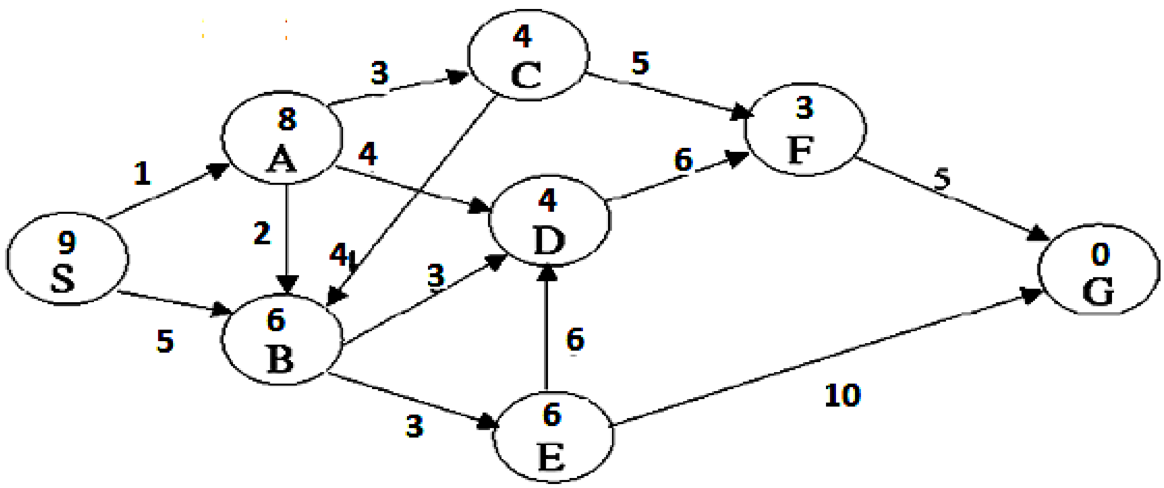
National University of Computer and Emerging Sciences, Lahore Campus

Course:	Artificial Intelligence		Course Code:	CS401
Program:	BS(Computer Science)		Semester:	Fall 2016
Duration:	60 min		Total Marks:	30
Paper Date:	28-02-2018		Weight	15%
Section:	D, E		Page(s):	6
Exam:	Mid I		Reg. No	

Name: ___________________________________ **Registration No** _______________________

Instruction/Notes:
Please read questions carefully, give reason/show all working for all questions
Answer all questions are on this question paper
Rough sheets can be used and attached at the end only if necessary.

Problem 1. Consider the search space shown in the figure below. In this problem the start state is **S**, and the goal state is **G**. The transition costs are next to the edges, and the heuristic estimate are given inside the nodes. Assume ties are always broken by choosing the state which comes first alphabetically



Part a) **[2 + 3 Points]**

i) Is the heuristic function admissible? (Give Reason)

ii) Is the heuristic function consistent? (Give Reason)

Part b) What is the order of states expanded and the path found by A* graph search algorithm when used on this problem? Also specify the nodes in the fringe/frontier/Queue when the goal is found. **[3 + 2 + 1 Points]**

Problem 2:

An intelligent but greedy automated agent/program is trying to solve the famous 0-1 knapsack problem. In this problem the agent has a set of items to choose from with each item having a **weight,** the agent has to carry if it picks that item, and a **value** the agent will gain by picking up the item**.** The agent has the capicity to carry only a **MAX_WEIGHT** while his job is to select a subset of items that **it can carry** such that the **total value gained** by the agent maximum.

The agent has been programmed by a slected team of students from a famous University (FAST) situated in the city of Lahore. The team programmed the agent such that it uses hill-climbing strategy to solve such optimization problems.

To represent state of the agent, while it is creating a plan to make a selection from a set of **n** items, the team used an array of size **n** with an index corresponding to one of the item in the set. Each index in the state array contains wither a 0 or a 1 where 0 at any index means that the agent will not pick the corresponding item and a 1 means the agent will pick the corresponding item.

Further, to generate successors of a state a very simple successor function is used that works by selecting an index and if the corresponding item is not already in the list of items to be picked it added it into the list by placing a 1 at the corresponding arry index.

Part a) If the ste of items contains 100 items **[1 + 1 + 3 Points]**

i. What is the size of search space the agent has to search from? (Justify)

ii. How many successors a state can have at max? (Justify)

iii. What will be a suitable evaluation function to carry out the hill-climbing search?

Part b) Now suppose that the set of items contain only **4** items. The weights of items and their value is given in the following table.

Item ID	**A**	**B**	**C**	**D**
Item Weight	**0.8 kg**	**0.7 kg**	**2 kg**	**3 kg**
Item Value	**Rs. 80**	**Rs. 70**	**Rs. 200**	**Rs. 320**

Table 1: Item Weights and Value

If the maximum capacity of the agent is **3.5 kg**. then what will be final solution found be the agent (Show Complete Working)

i) If starts in the following state **[2 Points]**

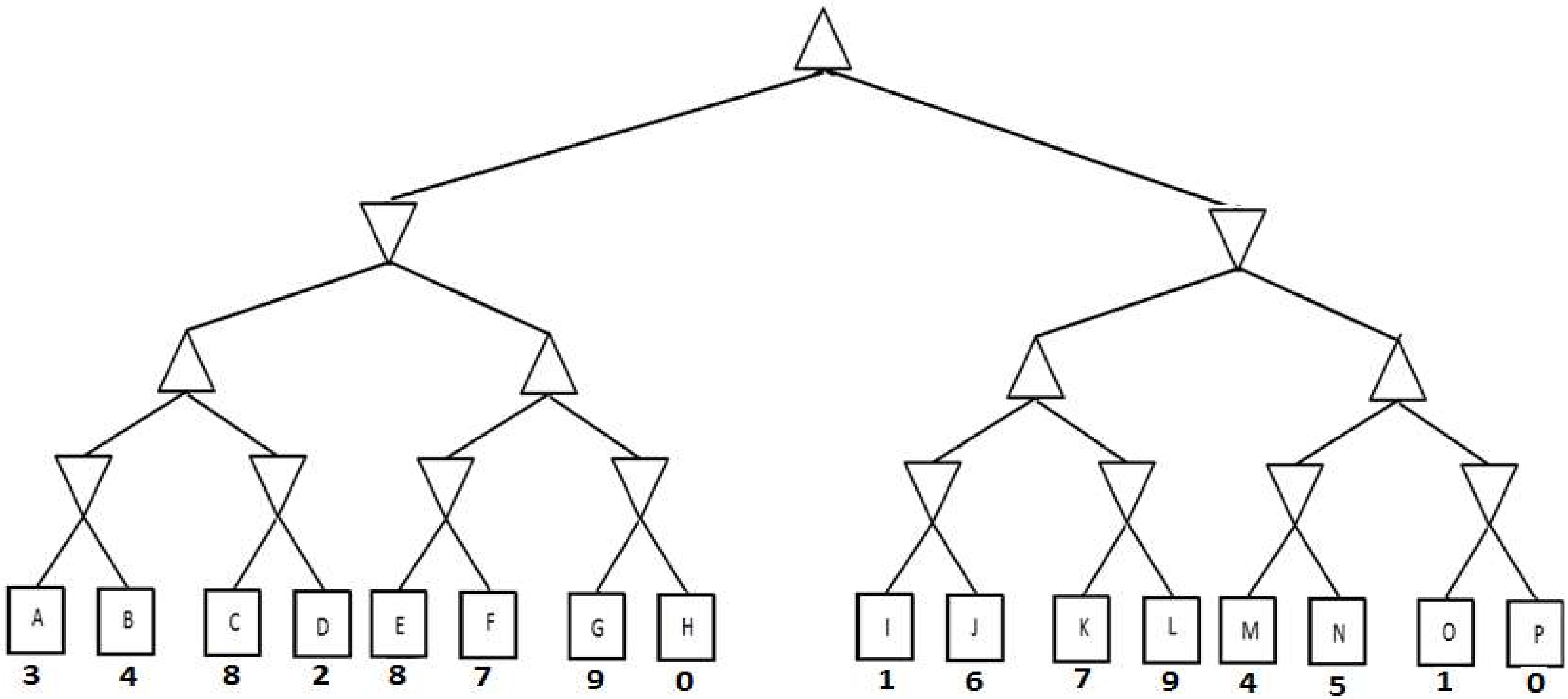
Initial State	**0**	**0**	**0**	**0**

ii) If starts in the following state **[3 Points]**

Initial State	**0**	**1**	**0**	**0**

Problem 3

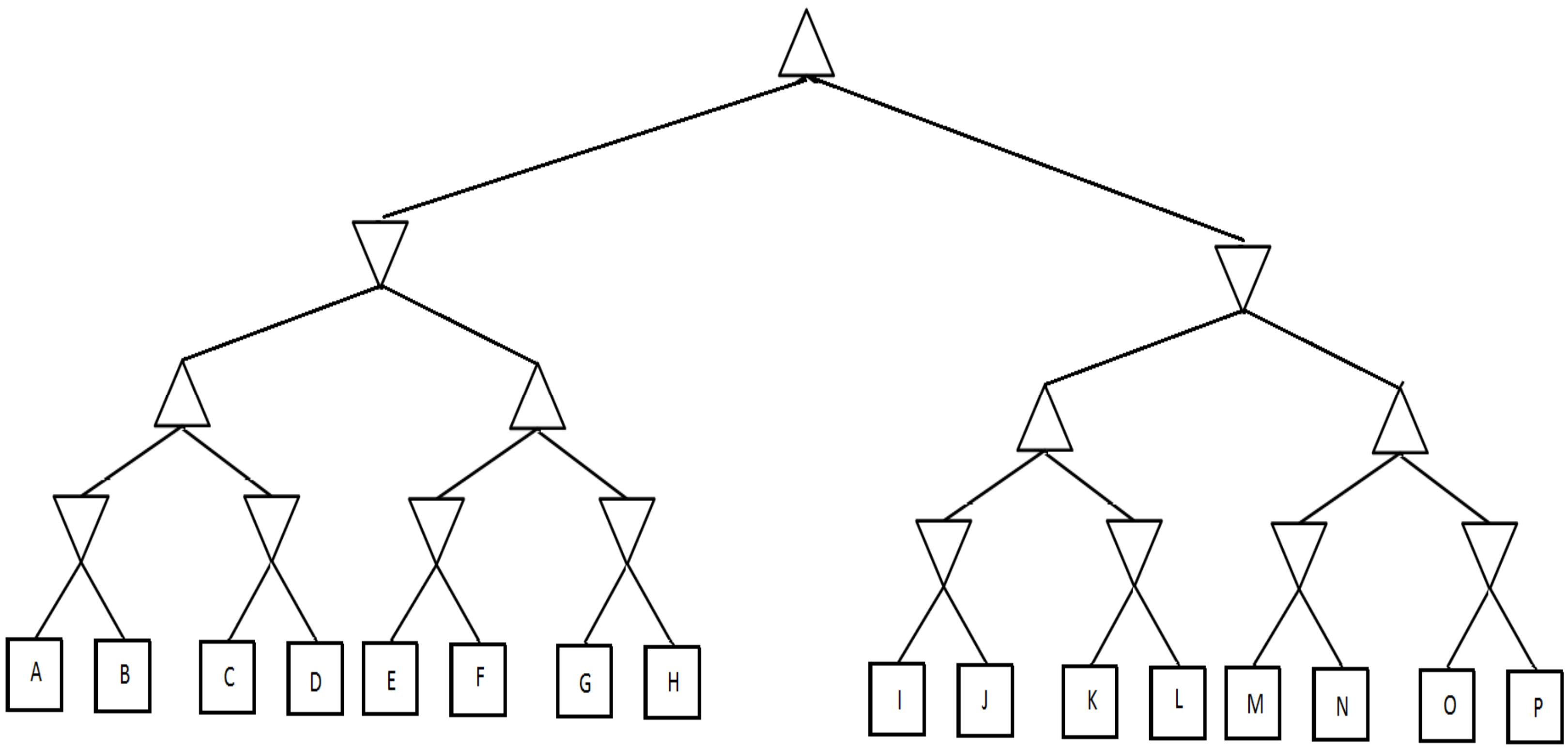
Figure on the following page shows a game tree of a hypothetical game. To find the best move of **MAX** (denoted by upwards pointing triangle) at the root, this game tree is to be searched using **minimax with α-β pruning**. Assume that the minimax search has been implemented such that the child/successors of any node are traversed from left to right order.



Part a) If value of each leaf node (specified as a square) is as given on the figure then compute the value of each intermediate node that will be assigned by minimax with prunning. On the figure also mark all branches that will be pruned by **α-β** prunning strategy and specify the corresponding values of **α and β** at the time of prunning **[5 + 5 Points]**

Bonus Part

Now assue that the values of the leaf nodes are not known and you are free to specify these values. Determin values of the nodes such that **maximun number of branch will be pruned** if minimax with **α-β pruning** is used to search the tree and children of a node are processesed from left to right. Also specify on the figure the branches that will be pruned. **[5 Points]**



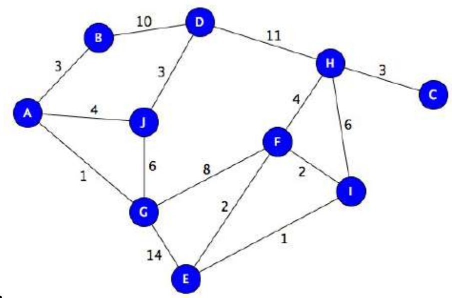
National University of Computer and Emerging Sciences, Lahore Campus

Course:	Artificial Intelligence	Course Code:	CS401
Program:	BS(Computer Science)	Semester:	Spring 2017
Duration:	60 Min	Total Marks:	30
Paper Date:	21-2-2017	Weight	15%
Section:	D and E	Page(s):	4
Exam:	Mid 1	Reg. No(Sec)	------------( )

Instruction/Notes: You can use rough sheet, but final working and answers should be on paper. You can use back side of paper as well.

Question 1: (10)

a) Consider the state space given in figure. Traverse a graph using uniform cost (graph) search to reach from start state A to goal state C. At each step show which node is explored and changes in frontier and explored list. Also maintain parent and g(n) for each node.



step 1

Frontier

Node	A
Cost g(n)	0
Parent	null

Explored

Node	
Cost g(n)	
Parent	

Step 2 — A is explored

Frontier

Node	B	J	G
Cost g(n)	3	4	1
Parent	A	A	A

Explored

Node	A
Cost g(n)	0
Parent	null

Step 3 — G is explored

Frontier

Node	B	J	F	E
Cost g(n)	3	4	9	15
Parent	A	A	G	G

Explored

Node	A	G
Cost g(n)	0	1
Parent	null	A

Step 4 — B is explored

Frontier

Node	J	F	E	D
Cost g(n)	4	9	15	13
Parent	A	G	G	B

Explored

Node	A	G	B
Cost g(n)	0	1	3
Parent	null	A	A

Step 5 — J is explored, cost and parent of D is updated

Frontier

Node	F	E	D
Cost g(n)	9	15	7
Parent	G	G	J

Explored

Node	A	G	B	J
Cost g(n)	0	1	3	4
Parent	null	A	A	A

Step 6 — D is explored

Frontier

Node	F	E	H
Cost g(n)	9	15	18
Parent	G	G	D

Explored

Node	A	G	B	J	D
Cost g(n)	0	1	3	4	7
Parent	null	A	A	A	J

Step 7 — F is explored, parent and cost of E and H is updated

Frontier

Node	E	H	I
Cost g(n)	11	13	11
Parent	G	D	F

Explored

Node	A	G	B	J	D	F
Cost g(n)	0	1	3	4	7	9
Parent	null	A	A	A	J	G

Step 8 — E is explored

Frontier

Node	H	I
Cost g(n)	13	11
Parent	D	F

Explored

Node	A	G	B	J	D	F	E
Cost g(n)	0	1	3	4	7	9	11
Parent	null	A	A	A	J	G	G

Step 9 — I is explored

Frontier

Node	H
Cost g(n)	13
Parent	D

Explored

Node	A	G	B	J	D	F	E	I
Cost g(n)	0	1	3	4	7	9	11	11
Parent	null	A	A	A	J	G	G	F

Step 10 — H is explored

Frontier

Node	C
Cost g(n)	16
Parent	H

Explored

Node	A	G	B	J	D	F	E	I	H
Cost g(n)	0	1	3	4	7	9	11	11	13
Parent	null	A	A	A	J	G	G	F	D

Step 11 — C is explored

Frontier

Node
Cost g(n)
Parent

Explored

Node	A	G	B	J	D	F	E	I	H	C
Cost g(n)	0	1	3	4	7	9	11	11	13	16
Parent	null	A	A	A	J	G	G	F	D	H

Step 12 — As C has been explored we can stop and back tracking will give us the following shortest path with path cost 16

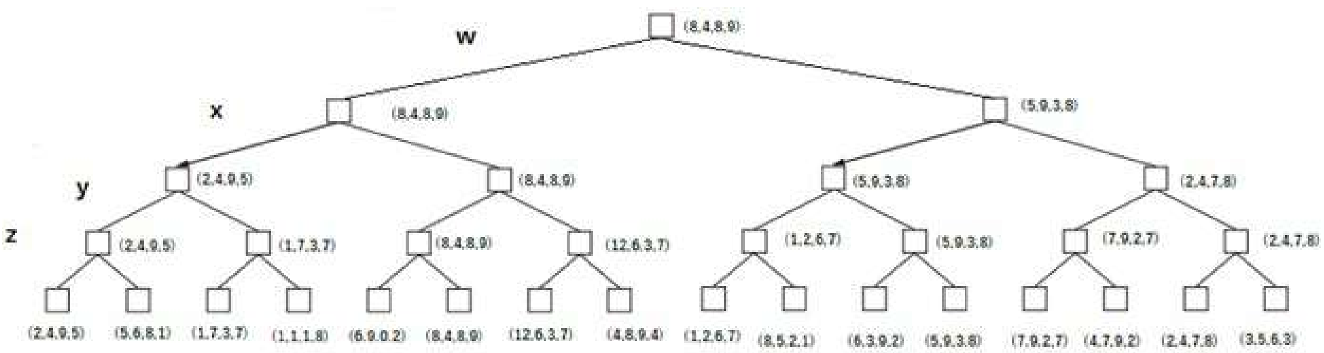
C <-- H <-- F <--G <--A

Figure 1 State Space

Roll No (Sec): ------------------------- ()

Question 3: (6 + 4)

a) Consider game tree of 4 player game (W, X, Y, Z). Terminal states show the Score of each player W, X, Y, Z . The game is played in partnership W and Y are partners, and X and Z partners. Each player is playing optimality and tries to maximize sum of its own and its partner's score. Partnership with highest score wins at the end of game. Find the node value of each node given in the tree. Which partners will win in the end of game if everyone plays optimally?



W and Y will win as scoreW+ScoreY is 16 and ScoreX+ ScoreZ is 13

b) Write a recursive function to get node value for part (a).

NodeValue (s) =

Utility if s is terminal node

argmax $_{\text{x=NodeValue (Result (s, a)) for all a } \epsilon \text{ actions(S)}}$ (x.score(s.player)+ x.score(s.partner))

Question 4: (10)

Consider the following Genetic Algorithm setup for some hypothetical problem

Population size= 6

Chromosome: array of 8 bits

Fitness function is given as f (n) = number of 1's in chromosomes

Goal: Fitness>= 6

Selection: Rank selection

Cross over method: One point, from random point.

Mutation Rate: 0%

Update method= population → 6 best from (population union population)

We have generated 6 random chromosomes in initial population given in table 2.

1. Find the fitness of chromosomes in initial population and their selection probability.
2. Perform first iteration using the selection, cross over and mutation method given above and generated new population.
3. Which chromosomes will go as population in 2nd iteration?

**NOTE: You can use suppose a random number if you need one, just mention its value where you use it. Show all steps and working clearly.

Table 1: Initial Population

		Fitness	Rank	Selection Probability%
Chromosome 1	00001101	3	6	22.2
Chromosome 2	11000000	2	2	7.4
Chromosome 3	00000000	0	1	3.7
Chromosome 4	10101000	3	6	22.2
Chromosome 5	00111000	3	6	22.2
Chromosome 6	01000011	3	6	22.2

Generating new population

C1: 00001101

C4: 10101000

Cross over point 4

New C1: 00001000

New C2: 10101101

C1: 00001101

C2: 11000000

Cross over point 1

New C3: 01000000

New C4: 10001101

C5: 00111000

C6: 01000011

Cross over point 4

New C5: 00110011

New C6: 01001000

New Chromosomes	Fitness
00001000	1
10101101	5
01000000	1
10001101	4
00110011	4
01001000	2

Chromosomes for second iteration	Fitness
01000011	3
10101101	5
01000011	3
10001101	4
00110011	4
00001101	3

National University of Computer and Emerging Sciences, Lahore Campus

	Course:	Artificial Intelligence	Course Code:	CS401
	Program:	BS(Computer Science)	Semester:	Fall 2016
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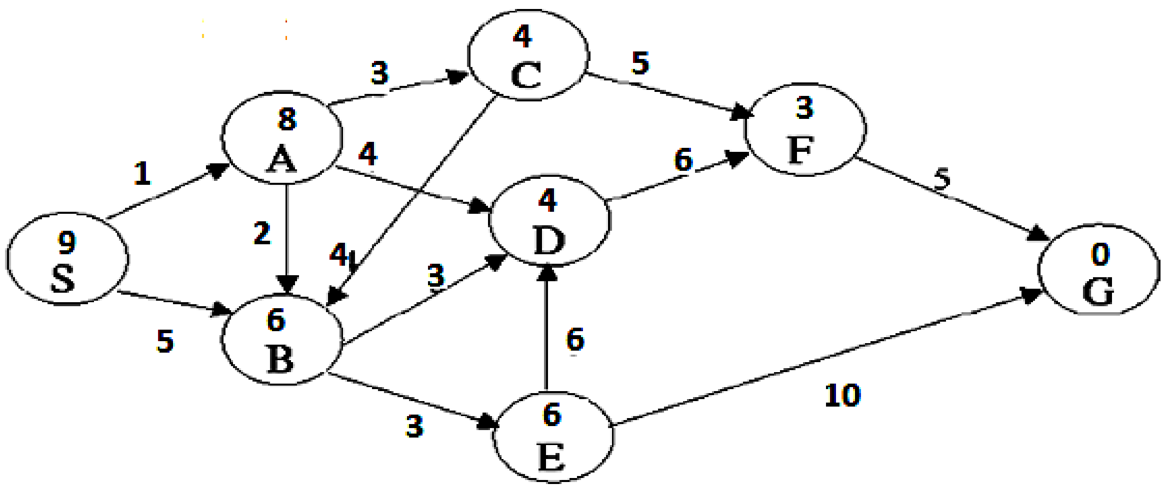

Part a) **[2 + 3 Points]**

i) Is the heuristic function admissible? (Give Reason)

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Initial State	**0**	**0**	**0**	**0**

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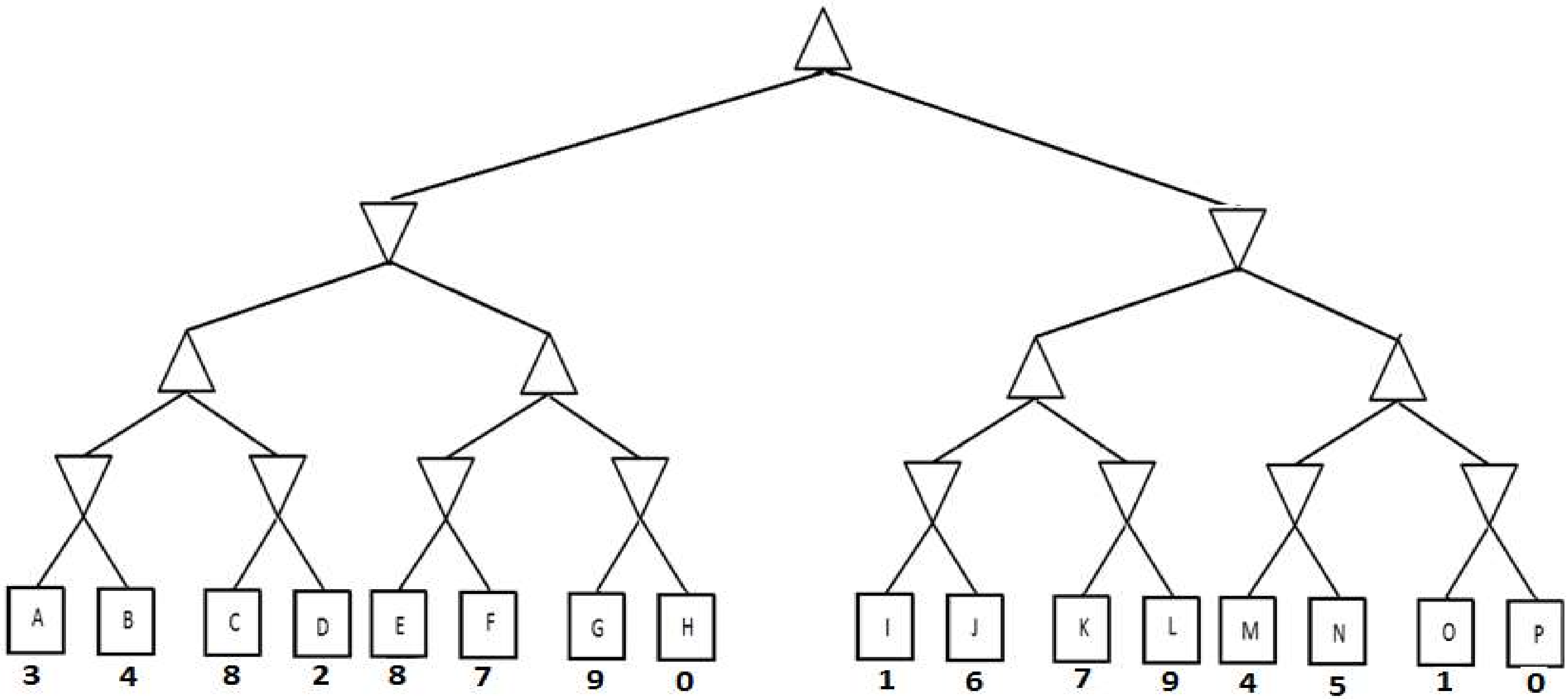
Initial State	**0**	**1**	**0**	**0**

Problem 3

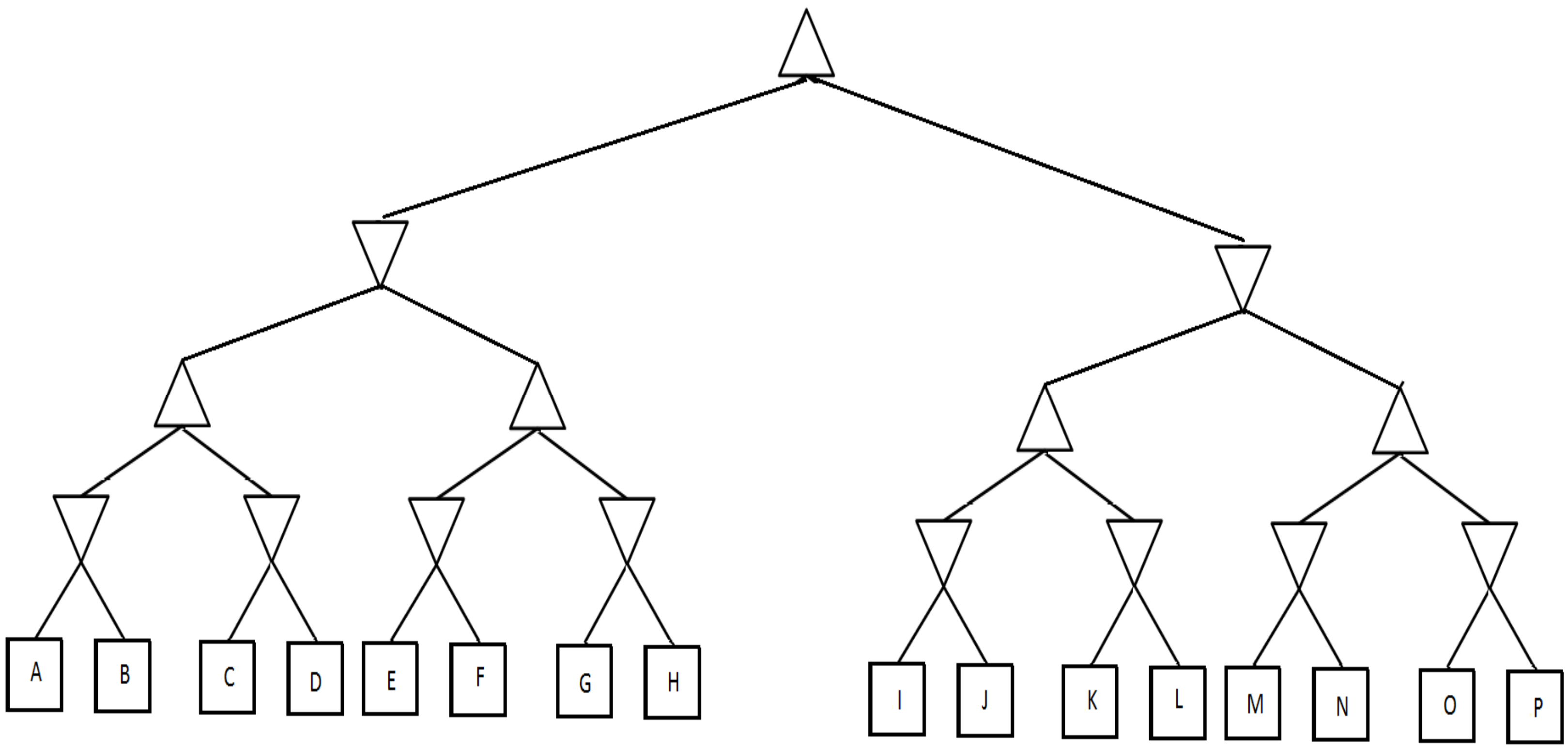
Figure on the following page shows a game tree of a hypothetical game. To find the best move of **MAX** (denoted by upwards pointing triangle) at the root, this game tree is to be searched using **minimax with α-β pruning**. Assume that the minimax search has been implemented such that the child/successors of any node are traversed from left to right order.



Part a) If value of each leaf node (specified as a square) is as given on the figure then compute the value of each intermediate node that will be assigned by minimax with prunning. On the figure also mark all branches that will be pruned by **α-β** prunning strategy and specify the corresponding values of **α and β** at the time of prunning **[5 + 5 Points]**

Bonus Part

Now assue that the values of the leaf nodes are not known and you are free to specify these values. Determin values of the nodes such that **maximun number of branch will be pruned** if minimax with **α-β pruning** is used to search the tree and children of a node are processesed from left to right. Also specify on the figure the branches that will be pruned. **[5 Points]**



National University of Computer and Emerging Sciences, Lahore Campus

	Course:	**Artificial Intelligence**	**Course Code:**	**CS401**
	Program:	**BS(Computer Science)**	**Semester:**	**Spring 2017**
	Duration:	**60 Min**	**Total Marks:**	**30**
	Paper Date:	**21-2-2018**	**Weight**	**15%**
	Section:	**D and E**	**Page(s):**	**4**
	Exam:	**Mid 1**	**Reg. No(Sec)**	**------------( )**

Instruction/Notes: You can use rough sheet, but final working and answers should be on paper. You can use back side of paper as well.

Question 1: (10)

a) Consider the state space given in figure. Traverse a graph using uniform cost (graph) search to reach from start state A to goal state C. At each step show which node is explored and changes in frontier and explored list. Also maintain parent and g(n) for each node.

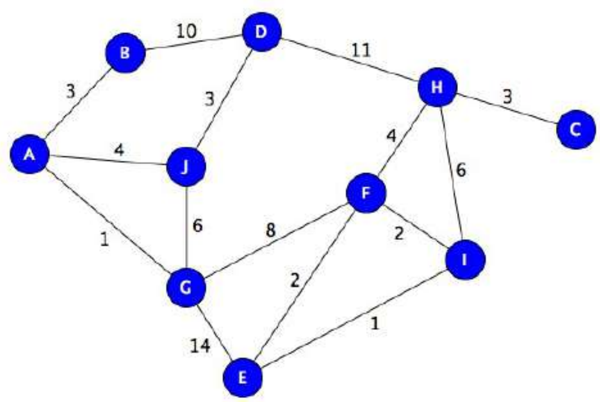
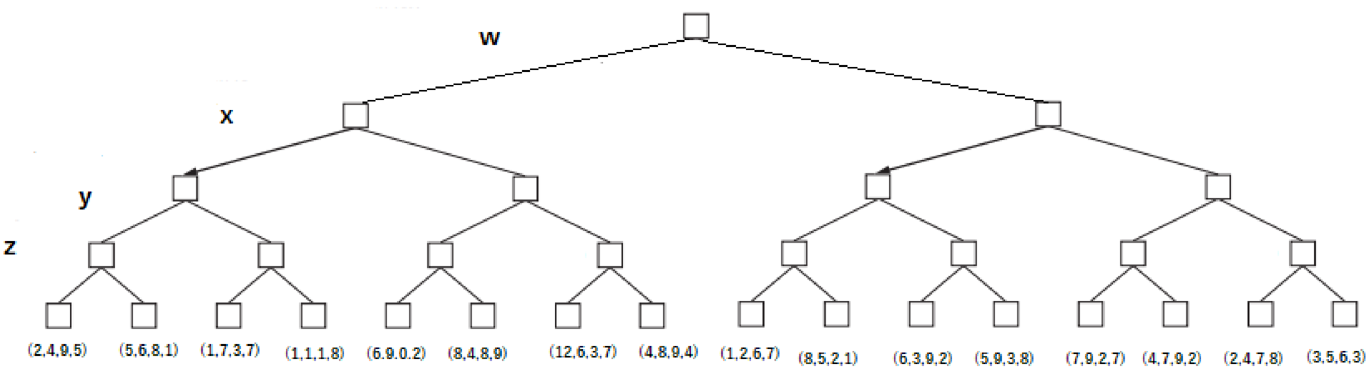


Figure 1 State Space

Question 3: (6 + 4)

a) Consider game tree of 4 player game (W, X, Y, Z). Terminal states show the Score of each player W, X, Y, Z . The game is played in partnership W and Y are partners, and X and Z partners. Each player is playing optimality and tries to maximize sum of its own and its partner's score. Partnership with highest score wins at the end of game. Find the node value of each node given in the tree. Which partners will win in the end of game if everyone plays optimally?



b) Write a recursive function to get node value for part (a).

Question 4: (10)

Consider the following Genetic Algorithm setup for some hypothetical problem

Population size= 6
Chromosome: array of 8 bits
Fitness function is given as f (n) = number of 1's in chromosomes
Goal: Fitness>= 6
Selection: Rank selection
Cross over method: One point, from random point.
Mutation Rate: 0%
Update method= population → 6 best from (population union population)

We have generated 6 random chromosomes in initial population given in table 2.

1. Find the fitness of chromosomes in initial population and their selection probability.
2. Perform first iteration using the selection, cross over and mutation method given above and generated new population.
3. Which chromosomes will go as population in 2nd iteration?

**NOTE: You can use suppose a random number if you need one, just mention its value where you use it. Show all steps and working clearly.

Table 1: Initial Population

Chromosome 1	00001101
Chromosome 2	11000000
Chromosome 3	00000000
Chromosome 4	10101000
Chromosome 5	00111000
Chromosome 6	01000011

Reg. No__________________ Section_____________

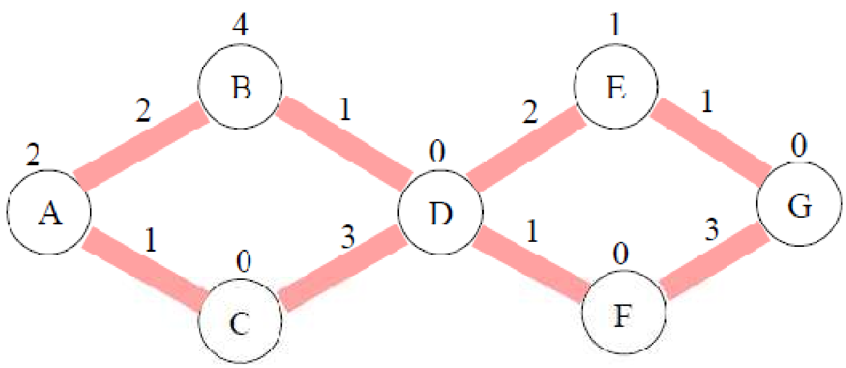
National University of Computer and Emerging Sciences, Lahore Campus

Course:	Artificial Intelligence	Course Code:	CS401
Program:	BS(Computer Science)	Semester:	Spring 2017
Duration:	60 Min	Total Marks:	25
Paper Date:	21-2-2018	Weight	12.5%
Section:	A, B and C	Page(s):	6
Exam:	Mid-I	Reg. No	------------

Instruction/Notes:
This is a closed book closed notes exam.
Calculators are allowed
Write your final answers on the provided space only.
Show your working on the back of these sheets or attach any extra sheets, containing your working, with this paper.
Write your ID and section on every page

Problem 1:

The following is a state-space search graph of a problem with nodes labeled using alphabets, edges are the thick shaded lines. The number above each edge giving the transition cost *(e.g., cost(C,D) = 3)* of the corresponding action and the number above each node is its heuristic value of that node *(e.g., h(A) = 2)*. The node labeled **A** is the initial state and that labeled as **G** is the goal state.



a) List, in order, the nodes that will be expanded if A* algorithm is used to find a path from the initial state to the goal state. [2 Points]

b) What path (if any) would be found by A* algorithm in part a. [2 Point]

c) What would be the state (nodes in it) of frontier/queue when the goal is found by A* algorithm in part b. [1 Point]

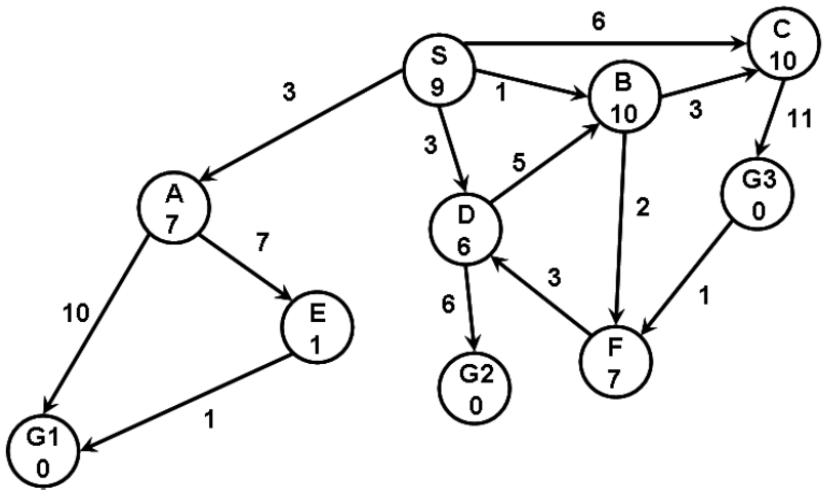
d) In one of my implementation of A* I made an interesting mistake. My implementation was identical to the correct A*, except that when it visits a node **n** that has already been expanded, it immediately skips **n** instead of checking if it needs to reinsert **n** into the priority queue. What path would be found by my erroneous implementation of A* in the search problem given in part a above. Also **comment** on the **effects of this change** on **completeness** and **optimality** of A* algorithm. [3 Points]

e) One FAST- student suggested an early termination strategy for A* algorithm. He suggested that A* must be stopped as soon as the first goal node G is found in some successor list instead of waiting until G is popped off the priority queue. What path would be found by his version of A*for the state-space search problem given in **part a**. Also **comment** on the **effects of this change** on **completeness** and **optimality** of A* algorithm. [3 Points]

f) Prove or disprove that the heuristic given in above problem is consistent. [1 Point]

Problem 2: **[1 +1 +1 +1 Points]**

For the state space search graph below, where **S** is the initial state and **G_1, G_2,** and **G_3** are goal states. Arcs are labeled with the cost of associated actions and the heuristic cost to a goal is shown inside the nodes. Mark the all goal states that might be returned by each of the following search algorithms.



DFS	€ **G_1**	€ **G_2**	€ **G_3**
BFS	€ **G_1**	€ **G_2**	€ **G_3**
Uniform Cost Search	€ **G_1**	€ **G_2**	€ **G_3**
A*	€ **G_1**	€ **G_2**	€ **G_3**

Problem 3: **[2 + 1 + 1 Points]**

a) If **h_1**, **h_2**, **h_3** and **h_4** is average of these four heuristics also admissible? Prove or disprove

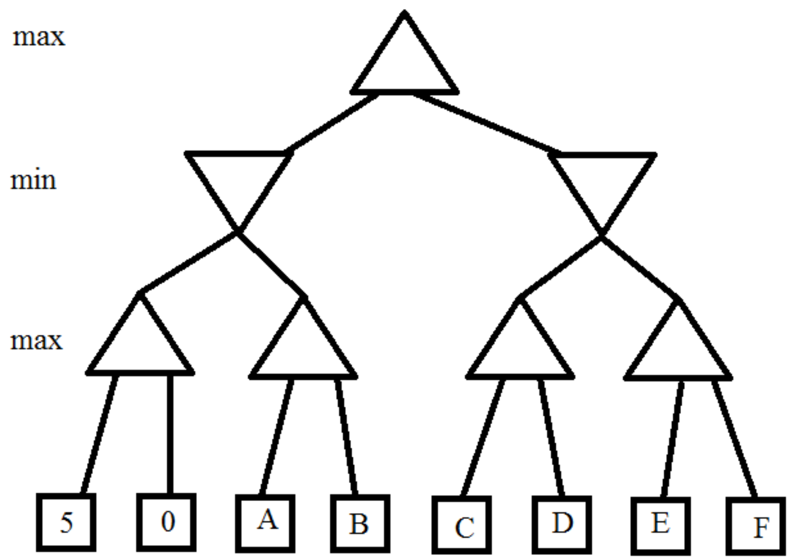
b) Which search algorithm is guaranteed to return optimal solution of step cost is a positive constant, but not otherwise?

c) Which of the following search algorithms are complete if goal is on finite depth and branching factor is also finite (chose all the apply)

€ DFS € BFS € Uniform Cost Search

Problem 4: **[1 + 1 + 1 + 2 Points]**

Consider the game tree picture below where A-F represent some real values. Assume the nodes are explored from left to right and standard alpha beta pruning is used.



a. Give a value of A such that B is pruned.

b. True or False: There are SOME values of A and B such that the sub-tree containing C and D is pruned?

c. Assuming that B = 5 and A = 5, give a value of C and D such that the sub-tree containing E and F is pruned.

d. If you are allowed to assign A-F arbitrarily, what is the MAXIMUM number of leaves that can be pruned? also specify the pruned leaves.

National University of Computer and Emerging Sciences, Lahore Campus

Course Name:	Artificial Intelligence	Course Code:	CS 401
Program:	BS(CS)	Semester:	Fall 2017
Duration:	1 hr	Total Points:	20
Paper Date:	Monday, 19 Sep 2017	Weight	10%
Section:	ALL	Page(s):	6
Exam Type:	Mid-I		

Student : Name:_______________________________ Roll No.________________ Section:_______

Instruction/Notes: Please Solve all questions on the question paper and also attach all rough sheets at the end.

Problem 1. Solving by Search

Classically the 15-Puzzle has been used to measure the performance of intelligent algorithms that solve a problem by searching for the optimal solution. This problem can be described as follows

The 15-puzzle is a sliding puzzle that consists of a 4 x 4 frame containing fifteen numbered square tiles, numbered from 1 to 15 and a missing tile as shown

The object of the puzzle is to place the tiles in order by making sliding moves that use the empty space.

Part a) Search Algorithms **[1 + 1 + 1 + 1 Points]**

Assuming that the **graph search version** of the blind search algorithms have been implemented and that on average 24 steps are needed to solve a puzzle.

What is the maximum and Minimum number of nodes/states explored by each of the following blind search algorithms on average? Give a brief description of your answer as well

Algorithm	Maximum	Minimum
DFS		
BFS		

Uniform Cost Search		
Iterative Deepening		

Part b) Search Algorithms **[3 + 1 Points]**

For the following **initial** and **goal** states, what is the maximum number of states/nodes expanded by A* and Greedy Best First Search if the **number of tiles out of place** is used as our heuristic value?

Initial State				Goal State			
1	2	3	4	1	2	3	4
5		7	8	5	6	7	8
10	6	11	12	9	10	11	12
9	13	14	15	13	14	15	

Part c) Search Algorithms **[2 Points]**

Also compute the maximum number of nodes expanded by BFS and DFS for this search problem.

Problem 2. [Short Questions] **[1 + 1 + 1 + 2 Points]**

Provide short answers (1-3 sentences) for each of the following questions.

a) In what way is iterative deepening is better than depth-first search? Use the four criteria typically used to compare the search algorithms.

b) Under what minimal conditions on the heuristic function the A* algorithm with graph-search is guaranteed to return an optimal solution?

c) State two major difference between Hill-climbing search and Best-first search.

d) What algorithm would result as a special case if
   i. Local beam search is applied with k = 1.

   ii. Genetic algorithm with population size N = 1 and a mutation rate of 1. (i.e. always apply mutation)

Problem 3: Game Playing: Four In A Row

Four in a Row is a two-player connection game in which the players first choose a color and then take turns dropping colored discs from the top into a seven-column, six-row vertically suspended grid. (As shown in figure 1a) The objective of the game is to be the first to form a horizontal, vertical, or diagonal line of four of one's own discs before your opponent.

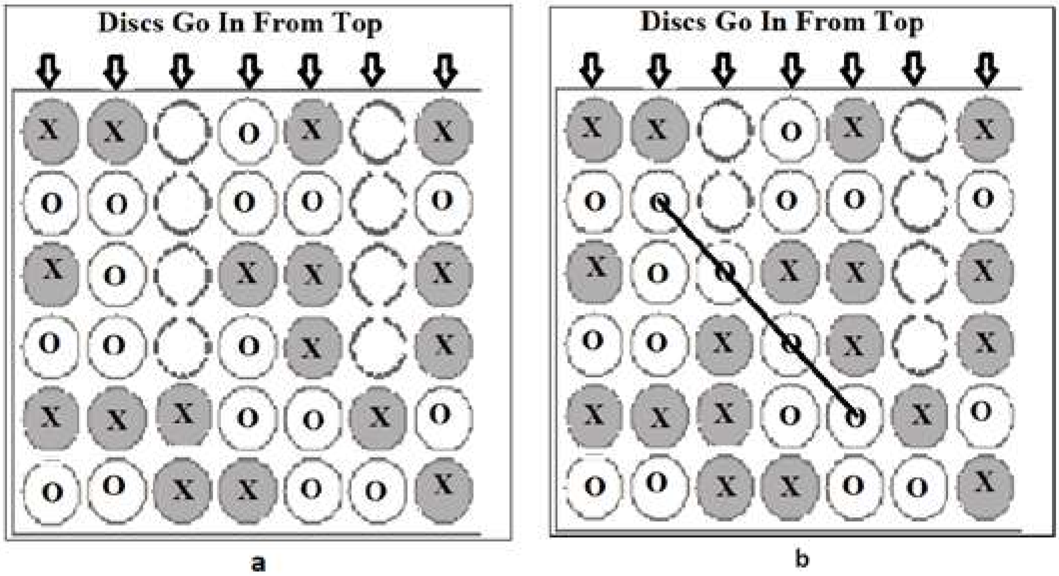


Figure: (a) Four in a Row Game grid. Discs can only be added from top. For sake of paper we represent two different colors by X and O (b) Grey Win state for, as four grey discs are on diagonal

The rules for Four in a Row are simple.

- The field (board) has seven columns and six rows.
- Two players play by alternately dropping a chip down one of the columns (from top).
- The chip drops to the lowest unoccupied spot in that column.
- The first player to get four of his own chips in a row, either vertical, horizontal, or diagonal, wins.
- The game ends in a draw if it fills before someone wins.

An AI student has decided to build an automatic player of FOUR IN A ROW using MINIMAX algorithm. Initially he decide to calculate a move at any given point in the game by building a complete game tree.

a) How many nodes will the game tree have when making the first move? (Give an approximate Answer) Note that at each level a player has about seven possible moves **[2 Points]**

The student figured out that the number of nodes in the game tree is large enough to prohibit building a complete game tree therefore he decided to choose a move by looking only **D** level deep in the tree. For this purpose he comes up with the following evaluation/expert function E.

```
int[][] evaluationTable = {{3, 4, 5,  7,  5,  4, 3},
                           {4, 6, 8,  10, 8,  6, 4},
                           {5, 8, 11, 13, 11, 8, 5},
                           {5, 8, 11, 13, 11, 8, 5},
                           {4, 6, 8,  10, 8,  6, 4},
                           {3, 4, 5,  7,  5,  4, 3}};
//This evaluation table is used as follows

int evaluateContent() {
        int utility = 128;
        int sum = 0;
        for (int i = 0; i < rows; i++)
            for (int j = 0; j <columns; j++)
                if (board[i][j] == 'O')
                    sum -= evaluationTable[i][j];
                else if (board[i][j] == 'X')
                    sum += evaluationTable[i][j];
        return utility + sum;
    }
```

The main idea behind this evaluation function is that the numbers in the table indicate the number of four connected positions which include that space. This gives a measurement of how useful each square is for winning the game and hence it helps decide the strategy. The student implemented the MINIMAX (with alpha-beta pruning) algorithm using his evaluation function. For the state of game given in figure 2**.**

b) It is X's turn to make a move show which move will be selected by the MINIMAX if D = 1

[3 Points]

BONUS) It is X's turn to make a move show which move will be selected by the MINIMAX if D = 2 [3 Points]

Department of Computer Science
National University of Computer and Emerging Sciences, Lahore Campus

Mid-I	**CS 401: Artificial Intelligence**	**Fall 2016**
Date: 9/21/16	**Marks: 30**	**Time: 60 Minutes**

Student ID ____________ **Name** ____________________ **Section** ____________

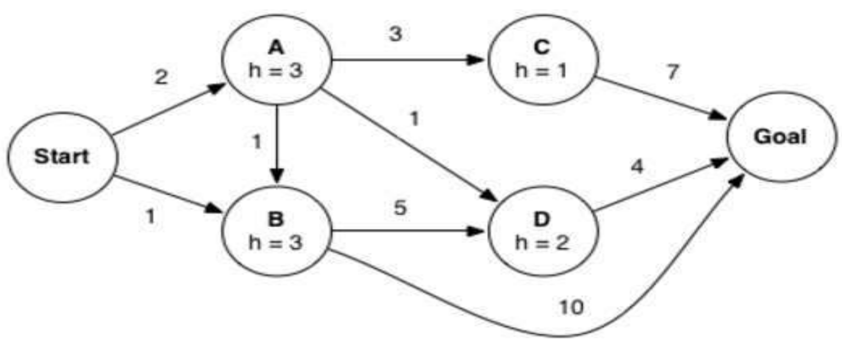
Notes:

This exam contains 7 pages (including this cover page) and 3 problems.
Check to see if any pages are missing. Enter all requested information on the top of this page, and put your ID on the top of every page, in case the pages become separated.
This a **closed Book, closed Notes** exam.

You are required to show your work on each problem on this exam.

1. (8 points) Consider A* graph search on the graph below. Arcs are labeled with action costs and states are labeled with heuristic values. Assume that ties are broken alphabetically (so a partial plan $S- > X- > A$ would be expanded before $S- > X- > B$ and $S- > A- > Z$ would be expanded before $S- > B- > A$.

   Note that to get full marks in each part you need to show complete algorithm working.



   (a) (2 points) In what order are states expanded by A* graph search?

(b) (2 points) What path does A* graph search return?

(c) (2 points) What is the final path returned by breadth-first graph search? Also specify the order of states expanded by BFS.

(d) (2 points) What is the final path returned by depth-first graph search? Also specify the order of states expanded by DFS.

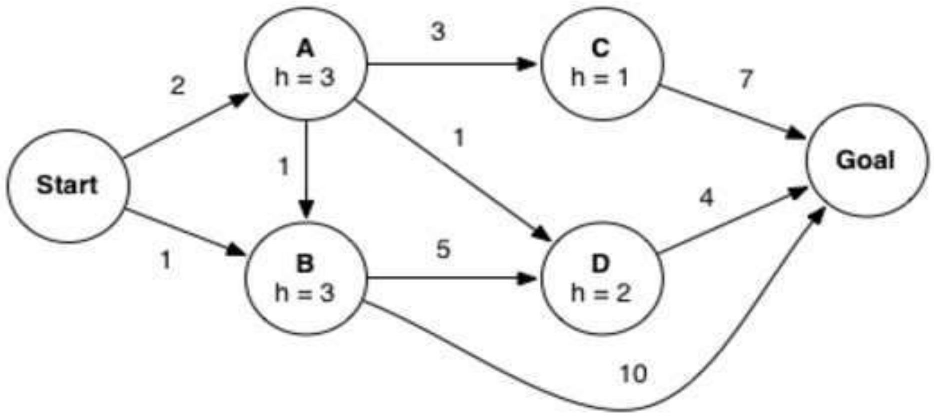
2. (10 points) The general algorithm for GRAPH-SEARCH as discussed in the class is given below.

```
function GRAPH-SEARCH(problem, fringe, strategy) return a solution, or failure
   closed ← an empty set
   fringe ← INSERT(MAKE-NODE(INITIAL-STATE[problem]), fringe)
   loop do
      if fringe is empty then return failure
      node ← REMOVE-FRONT(fringe, strategy)
      if GOAL-TEST(problem, STATE[node]) then return node
      if STATE[node] is not in closed then
         add STATE[node] to closed
         for child-node in EXPAND(STATE[node], problem) do
            fringe ← INSERT(child-node, fringe)
         end
   end
```

With the above implementation a node that reaches a goal state may sit on the fringe while the algorithm continues to search for a path that reaches a goal state. Let's consider altering the algorithm by testing whether a node reaches a goal state when inserting into the fringe. Concretely, we add the line of code highlighted below:

```
function EARLY-GOAL-CHECKING-GRAPH-SEARCH(problem, fringe, strategy) return a solution, or failure
   closed ← an empty set
   fringe ← INSERT(MAKE-NODE(INITIAL-STATE[problem]), fringe)
   loop do
      if fringe is empty then return failure
      node ← REMOVE-FRONT(fringe, strategy)
      if GOAL-TEST(problem, STATE[node]) then return node
      if STATE[node] is not in closed then
         add STATE[node] to closed
         for child-node in EXPAND(STATE[node], problem) do
            if GOAL-TEST(problem, STATE[child-node]) then return child-node
            fringe ← INSERT(child-node, fringe)
         end
   end
```

The modified algorithm can find a solution faster. However, this change might have affected some properties of the algorithm. To explore the possible differences, consider the graph of problem on also given below.



(a) (2 points) If using EARLY-GOAL-CHECKING-GRAPH-SEARCH with a Uniform Cost node expansion strategy, which path, if any, will the algorithm return?

(b) (2 points) If using EARLY-GOAL-CHECKING-GRAPH-SEARCH with an A* node expansion strategy, which path, if any, will the algorithm return?

(c) (3 points) Assume you run EARLY-GOAL-CHECKING-GRAPH-SEARCH with the Uniform Cost node expansion strategy. Which of the following statements are true. Justify your answer.

- The algorithm is complete

- The algorithm will return an optimal solution.

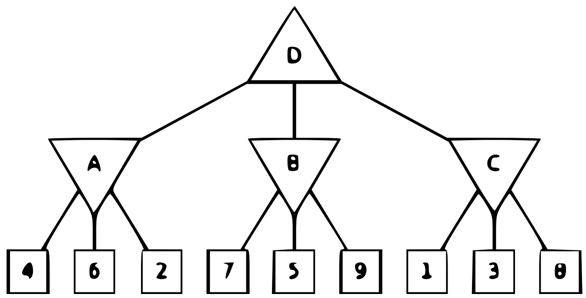
(d) (3 points) Assume you run EARLY-GOAL-CHECKING-GRAPH-SEARCH with the A* node expansion strategy and a consistent heuristic. Which of the following statements are true. Justify your answer.

- The algorithm is complete

- The algorithm will return an optimal solution.

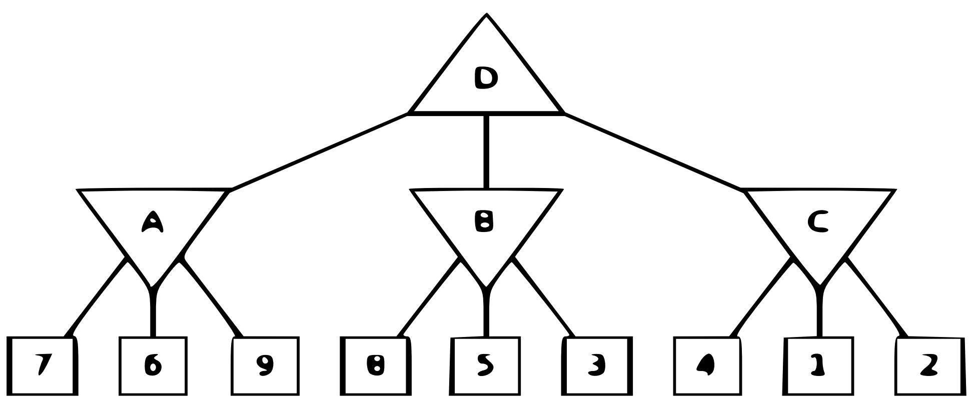
3. (12 points) **Minimax Search** For this problem you must use the given graphs to show your answers

   (a) (2 points) Consider the game tree shown below. Triangles that point up, such as at the top node (root), represent choices for the maximizing player; triangles that point down represent choices for the minimizing player. **Find the value of each node using minimax algorithm**.

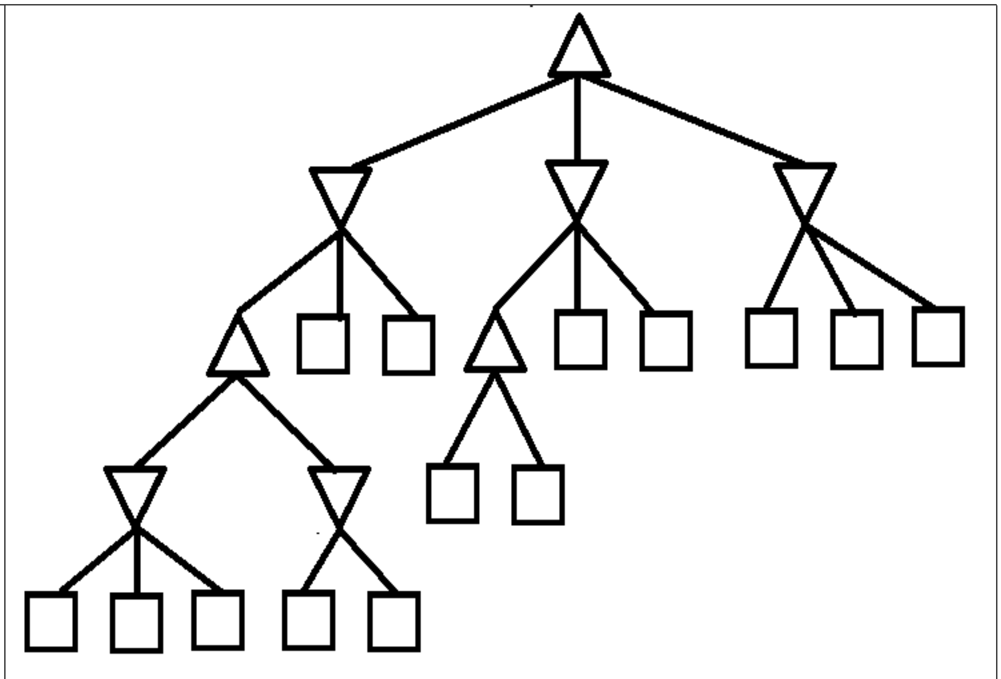


   (b) (4 points) Assuming both players in an adversarial game act optimally, use alpha-beta pruning to find the value of the root node with the game tree shown below.Assume that the search goes from left to right; when choosing which child to visit first, choose the left-most unvisited child.

   For each of the node specify the value computed by the algorithm. Also mark **an 'X'** for the nodes that don't get visited due to pruning. For



(c) (6 points) Now consider the two game trees shown below. For each of the graphs shown below specify the values of utility function for each of the terminal node (square nodes) that will cause the minimax algorithm prune the indicated branches. **Note that in this part every student will, typically, have a different answer**

No Pruning	
Pruning